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Designing Energy Tokens for Energy Communities

A Physically Grounded Governance Framework

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Designing Energy Tokens for Energy Communities: A Physically Grounded Governance Framework

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Abstract

Renewable energy communities (RECs) are playing a more important role in the transition towards decentralized and low-carbon energy systems. Existing research has extensively investigated energy management and benefit settlement within RECs. However, relatively limited attention has been devoted to governance mechanisms that encourage sustained community participation and equitable benefit distribution. Although energy token concepts have recently attracted growing interest, many of the existing approaches remain weakly connected to the physical operation of community energy systems.

This thesis proposes a physically grounded governance framework for designing and evaluating energy token mechanisms in renewable energy communities. The framework integrates a physical-layer energy community model with a governance-layer token mechanism through a sequential workflow. The physical layer models photovoltaic (PV) generation, optimizes PV and battery energy storage system (BESS) sizing and operational dispatch, and evaluates battery degradation and lifecycle techno-economic performance. The governance layer translates eligible renewable energy contributions into energy tokens and evaluates alternative governance designs involving token allocation, treasury management, token circulation, service redemption, and demand-response incentives. A multidimensional evaluation framework is further developed to systematically compare alternative governance configurations across multiple performance dimensions. The framework is demonstrated using a residential renewable energy community in Örebro, Sweden.

The results show that renewable energy contributions can provide a meaningful physical value anchor for token issuance while maintaining consistency between governance incentives and community energy operation. The analysis further demonstrates that governance performance is characterized by inherent trade-offs among renewable energy utilization, fairness, community activation, and token circulation. Through systematic evaluation of alternative governance configurations, the framework identifies governance strategies that balance these competing objectives under consistent physical operating conditions.

Overall, this thesis contributes a structured methodology for the systematic design and evaluation of energy token mechanisms in renewable energy communities. By grounding governance design in physical energy system operation, the proposed framework provides a practical foundation for future governance-oriented digital incentive mechanisms supporting decentralized energy systems.

Keywords: Renewable energy communities, Energy tokens, Community governance, Energy system modelling, Digital incentives

Sammanfattning

Lokala energigemenskaper spelar en allt viktigare roll i omställningen till decentraliserade och koldioxidsnåla energisystem. Tidigare forskning har i stor utsträckning fokuserat på energihantering och ekonomisk fördelning inom energigemenskaper, medan styrningsmekanismer som främjar långsiktigt deltagande och en rättvis fördelning av nyttor har fått betydligt mindre uppmärksamhet. Samtidigt har energitoken blivit ett växande forskningsområde, men många befintliga koncept har fortfarande en svag koppling till den fysiska driften av energisystem.

Denna avhandling presenterar ett fysiskt förankrat ramverk för utformning och utvärdering av energitokenmekanismer i förnybara energigemenskaper. Ramverket integrerar en fysisk modell av energigemenskapen med en styrningsnivå baserad på energitoken genom ett sekventiellt arbetsflöde. Den fysiska modellen omfattar solelproduktion, optimering av dimensionering och drift av batterilagringssystem samt utvärdering av batteridegradering och teknoekonomisk prestanda ur ett livscykelperspektiv. Styrningsnivån omvandlar berättigade bidrag av förnybar energi till energitoken och utvärderar alternativa styrningsstrategier som omfattar tokenfördelning, treasury-förvaltning, tokencirkulation, inlösen av tjänster och incitament för efterfrågeflexibilitet. Ett flerdimensionellt utvärderingsramverk utvecklas dessutom för att systematiskt jämföra alternativa styrningskonfigurationer utifrån flera prestationsdimensioner. Ramverket demonstreras med hjälp av en bostadsbaserad energigemenskap i Örebro, Sverige.

Resultaten visar att bidrag av förnybar energi kan utgöra en meningsfull fysisk värdeförankring för utfärdandet av energitoken samtidigt som styrningsincitamenten förblir förenliga med energigemenskapens faktiska drift. Analysen visar vidare att styrningsprestanda kännetecknas av inneboende avvägningar mellan utnyttjande av förnybar energi, rättvisa, aktivering av gemenskapen och tokencirkulation. Genom en systematisk utvärdering av alternativa styrningskonfigurationer identifierar ramverket styrningsstrategier som balanserar dessa konkurrerande mål under konsekventa fysiska driftförhållanden.

Sammanfattningsvis bidrar denna avhandling med en strukturerad metodik för systematisk utformning och utvärdering av energitokenmekanismer i förnybara energigemenskaper. Genom att förankra styrningsdesignen i den fysiska driften av energisystem erbjuder det föreslagna ramverket en praktisk grund för framtida styrningsorienterade digitala incitamentsmekanismer som stödjer decentraliserade energisystem.

Nyckelord: Förnybara energigemenskaper, Energitoken, Gemenskapsstyrning, Energisystemmodellering, Digitala incitament

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I hope that the work presented in this thesis can contribute to the continued development of renewable energy communities by providing a practical foundation for future governance research. I also hope that the proposed framework will continue to evolve alongside future research and practical implementation as energy communities become increasingly widespread.

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List of Abbreviations

BESS	Battery Energy Storage System
BTM	Behind-the-Meter
CAPEX	Capital Expenditure
CEC	Citizen Energy Community
DSO	Distribution System Operator
DR	Demand Response
EC	Energy Community
EFC	Equivalent Full Cycle
EMD	Electricity Market Directive
EU	European Union
FIFO	First-In, First-Out
GDPR	General Data Protection Regulation
IRR	Internal Rate of Return
KPI	Key Performance Indicator
LP	Linear Programming
MiCA	Markets in Crypto-Assets Regulation
MILP	Mixed-Integer Linear Programming
NPV	Net Present Value
P2P	Peer-to-Peer
PSD2	Revised Payment Services Directive
PV	Photovoltaic
REC	Renewable Energy Community
RED II	Renewable Energy Directive (Recast)
SCR	Self-Consumption Rate
SME	Small and Medium-sized Enterprise
SOC	State of Charge
SSR	Self-Sufficiency Rate

1 Introduction

1.1 Background and Motivation

The global energy sector is in the transition towards greater use of renewable and low-carbon energy resources as a replacement of traditional fossil fuels [1]. This shift drives the reshaping of energy system architecture, from centralized and continuous energy generation systems to decentralized and intermittent energy conversion sites [2]. However, the variability of renewable energy generation, especially solar and wind, brings new challenges to grid stability. The mismatch between energy production and demand makes the efficient utilization of renewable energy more complex, which calls for innovative mechanisms to better coordinate the generation and consumption within the energy system [3].

Under this context, it has become increasingly important to improve the coordination of local energy balance, which helps to reduce local peak loads, ease grid congestion, and optimize the utilization of renewable energy resources. Energy communities (ECs), broadly understood as collective initiatives through which local actors jointly organize and manage energy-related activities for shared benefits, have therefore emerged as a promising organizational model to facilitate such coordination [4]. In the European Union (EU), the concept of energy communities has gained increasing regulatory recognition through the Clean Energy Package, a set of legislative measures that formally recognizes citizen participation and grants energy communities the right to generate, consume, store, and share electricity collectively [5]. ECs are therefore getting more attention as important actors in enabling decentralized and participatory energy transitions.

Despite the potential benefits, energy communities still face significant coordination challenges [6]. The spatio-temporal mismatch of electricity generation and consumption remains persistent within decentralized energy systems. In communities with high penetration of distributed PV systems, it occurs frequently that electricity generation exceeds the community's immediate demand. During such periods, surplus electricity is often exported to the grid at relatively low economic value. However, the community may still rely on grid electricity when local generation is insufficient. This brings in a key challenge of effectively coordinating the utilization of surplus electricity within energy communities.

Several approaches such as fixed energy sharing rules or peer-to-peer (P2P) electricity trading have been proposed to address this issue [5]. However, these mechanisms may involve complex market structures, regulatory constraints, or inflexible allocation schemes that ignore the extent of individual contributions. Growing interest has been observed in alternative coordination mechanisms that can represent

energy contributions more flexibly and support more transparent governance of shared resources.

With the increasing digitalization of energy systems, new possibilities have been explored. In particular, token-based mechanisms are getting more popular as potential tools for representing value and individual contributions within distributed energy systems [7]. In energy communities, tokens can function as digital units that represent either the value of locally generated electricity or the contribution of individual participants to the collective energy system. Many early discussions about tokenization of energy focused more on tokens as alternative media of exchange or settlement instruments within the local energy markets [8]. In principle, locally generated electricity can serve as a unit of value to facilitate exchanges within an energy community. However, electricity itself cannot be a direct exchange medium due to spatio-temporal value variance and physical constraints of electricity grid. Therefore, its direct use as a transferable exchange medium is limited and tokenization is increasingly being explored as a socio-technical governance mechanisms [9]. Tokens can provide a programmable representation layer through which selected physical energy contributions are encoded into digital units according to predefined rules. Their issuance and allocation do not necessarily correspond one-to-one with total electricity generation, but may depend on the specific objectives and eligibility criteria of the token design. This programmability enables communities to implement more advanced governance strategies, such as customized token issuance, fair resource allocation, and demand-response incentives.

The practical design of these token-based mechanisms for residential energy communities remains relatively underexplored. While previous studies have examined the applications of energy tokens to renewable energy incentives, local trading, and socio-economic participation in communities, limited attention has been paid to the development of an integrated framework, which explicitly connects physical energy system dynamics with token-based governance and incentive mechanisms [10].

To address this gap, this thesis investigates how token-based governance mechanisms can be linked to the physical operation of residential energy communities to represent surplus electricity contributions and support the coordination of shared community resources.

1.2 Research Objectives and Questions

This thesis aims to design and evaluate a token-based governance framework that links physical energy flows with digital incentive structures. It explores how surplus electricity generated in a residential energy community can be represented through energy tokens and used as a coordination tool for shared community resources. To achieve this goal, the research is structured around four key dimensions.

Firstly, the study develops a physical energy model to characterize the generation, consumption, storage, and sharing of electricity within residential energy communities. Different physical and operational scenarios are evaluated to quantify surplus electricity generation and community energy flows, namely a photovoltaic-only (PV-only) scenario, a photovoltaic plus battery energy storage system (PV-BESS) scenario, and a photovoltaic plus battery energy storage system with a behind-the-meter charging strategy (PV-BESS-BTM) scenario.

Secondly, a token-based governance framework is designed based on the generated physical energy flows. Surplus electricity contributions are translated into digital tokens, with token issuance linked to the utilization of locally generated surplus electricity within the community. These tokens are further integrated with a governance layer which comprises benefit allocation rules, demand response incentives, service utilization, and token supply control.

Thirdly, alternative energy community configurations, combining different physical scenarios and token governance settings, are compared under a multidimensional evaluation framework. The assessment considers activation, economic value, fairness, and token persistence in order to identify effective token design strategies and potential trade-offs among different configuration choices.

Finally, the regulatory context of token-based governance systems is examined. The analysis considers the EU regulatory framework and the Swedish context to identify legal constraints and design boundaries relevant to the potential implementation of energy tokens.

Accordingly, the following research questions are addressed:

1) How can surplus electricity contributions be translated into digital tokens in residential energy communities?
2) How do different token-based governance mechanisms influence the coordination of surplus electricity and shared community resources?
3) Which token design configurations provide a favorable balance across activation, fairness, economic value, and token persistence?
4) What regulatory constraints shape the feasible design space of token-based governance systems within the EU and Swedish context?

1.3 Scope and Limitations

This thesis focuses on the design and evaluation of token-based governance mechanisms for residential energy communities. The study investigates how digital tokens can represent locally generated surplus electricity and support the coordination of shared community resources.

The research is organized in an integrated modelling framework which links physical energy flows with a token-based governance mechanism. A Swedish residential energy community consisting of multiple buildings equipped with rooftop PV systems is selected as the reference case study. The Swedish context is particularly relevant because its national framework for implementing EU energy-community provisions is still evolving. Three community energy configuration scenarios are examined: PV-only scenario, PV-BESS scenario, and PV-BESS-BTM scenario. These scenarios allow for the analysis of how different physical and operational configurations change the utilization of surplus electricity and further influence the performance of token-based governance mechanisms.

Within the modelling framework, surplus electricity contributions are translated into energy tokens and integrated with community governance functions, such as benefit allocation, demand response incentives, service utilization, and token supply control. The study focuses on community-level coordination and governance rather than the implementation of a fully operational decentralized electricity market or bilateral P2P trading schemes.

In addition to the technical, economic, and governance performance considered within the modelling framework, the broader sustainability implications of the proposed approach are discussed from environmental, social, and economic perspectives. The study does not aim to provide a comprehensive sustainability assessment, and lifecycle environmental impacts or broader societal impacts are outside the defined analytical scope.

Several limitations should be acknowledged. The energy system analysis relies on reconstructed datasets rather than fully operational real-world community datasets, particularly for the modelling of BESS integration. In addition, token governance mechanisms are evaluated using rule-based simulations to identify the coordination potential, rather than modelling behavioral responses or adaptive decision-making by individual community members. Also, the evaluation is conducted within a single case study context, which limits the generalizability of the findings across different community characteristics, regulatory environments, or market conditions. Finally, the service utilization layer is represented through a simplified service-credit mechanism and does not model detailed service markets, service providers, or transaction dynamics.

2 Literature Review and Conceptual Foundation

Building on the research objectives, questions, and scope established in Chapter 1, this chapter further reviews the conceptual and empirical foundations of token-based governance mechanisms for residential energy communities. The review is organized around five interconnected themes: energy communities and local energy coordination, energy flow modelling in energy communities, tokenization in energy systems, existing energy token projects, and regulatory context governing energy communities and digital tokens.

Through this structure, the chapter first explains the emergence of energy communities and the coordination challenges related to surplus electricity and shared community resources. Then, it reviews the existing approaches for modeling energy flows in energy communities, highlighting their roles as the foundation for token design. The concept of tokenization is further discussed in terms of representation, incentives, and governance, followed by a comparative analysis of energy projects. The regulatory environment is then analyzed to identify the legal boundaries that shape the feasible design space of energy-token mechanisms. Finally, Section 2.6 synthesizes the findings from these interconnected research streams, identifies the research gaps addressed in this thesis, and derives the analytical implications that guide the methodology developed in Chapter 3.

2.1 Energy Communities and Local Energy Coordination

2.1.1 Emergence of Energy Communities

The global energy sector is undergoing a profound transformation at an exponential rate. Different from a top-down energy value chain characterized by centralized production and rigid distribution network, it gradually shifts to a collaborative ecosystem of self-managed prosumers who own distributed energy resources and can act independently on liberalized energy markets [6]. Energy communities are proposed as an effective tool to build a more flexible and decentralized energy systems in which citizens can be an active part of this energy transition paradigm [11].

An energy community is typically composed of a group of households or local actors, which often generate, consume, store, and share electricity collectively through distributed renewable energy systems such as rooftop PV. Multifaceted benefits of energy communities have been studied [12]. Technologically, ECs help to enhance grid flexibility and decarbonization by integrating distributed energy resources and storage as a single entity. Economically, the members' costs can be lowered down through optimization of local generation and consumption matching and demand-side management. Socially, ECs promote the catalysis of behavioral engagement to shift

consumption habits, mobilize private investment, and address energy poverty. They are the key supportive factors in a just and resilient energy transition.

Through the adoption of the Clean Energy for all Europeans package in 2019, the EU for the first time introduced the concept of energy communities into its legislation [13]. Two types of energy communities are defined in the legal framework, namely renewable energy communities (RECs) introduced in Revised Renewable Energy Directive (RED II) 2018/2001 [14] and citizen energy communities (CECs) defined in Electricity Market Directive (EMD) 2019/944 [15]. Table 2.1 summarizes and compares their main characteristics and differences as follows [6], [11], [16].

Table 2.1: Comparison of RECs and CECs

Energy Communities	**RECs**	**CECs**
Form	A legal entity (RED II, Article 2, para 16)	A legal entity (EMD, Article 2, para 11)
Primary Purpose	Providing **environmental, economic or social community benefits** rather than focusing on financial profits.	
Membership Eligibility	In accordance with the applicable national law, it is based on **open and voluntary participation**, **autonomous**, and effectively controlled by shareholders or members located in the proximity of the renewable energy projects that are owned and developed by that legal entity; the shareholders or members of which are **natural persons, SMEs or local authorities, including municipalities.**	It is based on **voluntary and open participation** and effectively controlled by members or shareholders that are **natural persons, local authorities, including municipalities, or small enterprises.**
Technological Activities	RECs shall be entitled to **produce, consume, store and sell renewable energy, including through renewable power purchase agreements.**	CECs may engage in **generation**, including from renewable sources, **distribution, supply, consumption, aggregation, energy storage, energy efficiency services or charging services for electric vehicles or provide other energy services** to its members or shareholders.

Generation Technology	Any type of **renewable energy technology** can be included in RECs, such as PV, wind, hydro, biomass, etc.	CECs operate in the electricity sector and are **technology-neutral** (either fossil fuels or renewable energy)

RECs are specifically centered on renewable energy and require participating members to be located in proximity to the relevant projects. In contrast, CECs cover a broader range of electricity-sector activities and are not subject to the same geographical proximity requirement, although both models prioritize environmental, economic, or social community benefits over financial profit. Together, these two definitions have set the basic regulatory framework for ECs in the EU. There are more than 8,000 energy communities existing across the EU according to the 10th State of the Energy Union report published in November 2025 [17].

In Sweden, energy communities are also viewed as a crucial element of the sustainable energy transition. However, there are currently no laws or regulations specifically designed to promote the development of ECs. Instead, the general support measures aim at promoting new energy infrastructures or energy efficiency. For example, the Swedish government has applied an installation tax credit for solar panels and wind power, as well as tax reductions for small-scale producers [18]. In response to the EU's Clean Energy package from 2019, the Swedish Energy Market Inspectorate (Ei) has been responsible for investigating and proposing recommendations to translate related EU directives into national legislation [19].

2.1.2 Energy Surplus and Coordination Challenges

Despite their theoretical potential, energy communities often face operational challenges in practice due to the spatio-temporal mismatch between local electricity generation and consumption. More specifically, distributed renewable energy systems like rooftop PVs normally produce electricity intermittently depending on weather conditions and time of the day. In contrast, households consume electricity based on their daily routines and consumption behaviors, leading to different temporal demand patterns. As a result, periods of electricity surplus or deficit may occur frequently. Without effective energy storage measures or flexible demand management, surplus electricity is normally exported to the external grid at lower economic values while households still rely on relatively expensive grid imports during periods of low renewable energy generation.

Figure 2.1 illustrates a typical energy system structure in residential energy communities, which further reveals the complexity of local energy coordination.

Households equipped with rooftop PV systems generate electricity locally and remain connected to the external electricity grid. The electricity generated within the community can be used directly by households, stored in a community-level BESS for later use, or fed into the grid. When local generation is insufficient, the electricity can be supplied either from the shared BESS or from the grid. This configuration creates a multi-directional energy flow structure, which involves households, shared storage infrastructure, and the external grid network. In reality, the structure can be more complex depending on the composition of community members, the energy storage measures, the grid conditions, and so on.

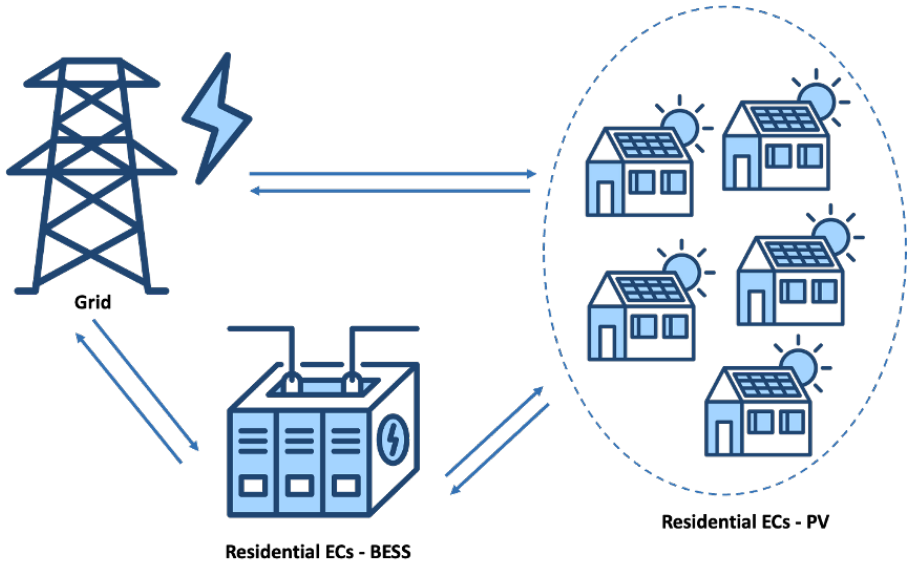


Figure 2.1: Typical energy system structure of residential energy communities

Although the integration of community energy storage and more advanced metering and distribution infrastructures can provide additional flexibility for managing renewable energy variability, the difficulty of coordination is positively correlated with the complexity of the physical energy system. Multiple households participate in electricity generation, and each may benefit differently from shared resources within the community. Therefore, how to allocate, store and utilize surplus electricity become a collective coordination issue. If the coordination mechanisms are not appropriately designed, the potential benefits of local renewable energy generation and shared infrastructure may not be fully realized. Consequently, various mechanisms have been proposed to improve the coordination of generation and consumption within energy communities, which will be discussed in the following section.

2.1.3 Existing Coordination Mechanisms

Plenty of coordination mechanisms have been discussed to address the coordination challenges associated with surplus electricity and shared resources in energy communities. These mechanisms can be generally categorized into three main types: rule-based coordination, market-based coordination, and centralized coordination. Each type presents distinct advantages and operational limitations. Within energy communities, these approaches are normally implemented through specific mechanism, such as fixed energy-sharing rules, market-based peer-to-peer trading, and centralized community energy management systems.

Fixed energy sharing mechanisms represent one of the most common implementations of rule-based coordination. As the name suggests, fixed rule-based energy sharing mechanism applies static or fixed energy sharing rules, such as proportional or equal sharing. They are widely adopted in early pilot projects due to their simplicity. Although these rules are easy to implement and align with the current regulatory billing systems, they are highly inflexible dealing with dynamic energy demands. Minuto and Lanzini (2022) have demonstrated that when community profits are distributed equally based on the shares of ownership, the most critical drawback is the absence of feedback on the member's demand behavior [20]. This allocation approach fails to promote virtuous energy consumption. Verschae et al. (2016) criticize this approach for erroneously assuming uniform user reactions while neglecting the crucial trade-off between power control and individual quality of life [21]. Tushar et al. (2020) also emphasize, if energy management techniques are not highly prosumer centric and fail to demonstrate tangible and individualized benefits, prosumers may decide to disengage entirely or go off grid [22]. This brings inefficient outcomes to both the grid and the community.

Market-based P2P energy trading mechanisms aim to address incentive issues through direct economic exchange between community members. Even though P2P markets offer high flexibility and individual autonomy, they face significant real-world barriers. From a regulatory perspective, local energy sharing is often hindered by rigid market structures designed for large-scale players. Vardanyan (2025) presents this situation within the Swedish context, where citizens are not economically motivated to engage in P2P trading due to a monopolized grid and the lack of a legal framework to incentivize coordination between distribution system operators and the energy community [18]. It happens even in countries like France, which have made progress in the Collective Self-Consumption (CSC) legislation. Strict operational boundaries, including physical distance limits of 2 kilometers and cumulative capacity caps of 3 MW, restrict the community's ability to achieve economies of scale [23]. Moreover, at the physical layer, unchecked P2P energy transfer can exacerbate grid stress, leading to reverse power flows and voltage violations in low-voltage networks [24]. Recently, decentralized ledgers such as blockchain are frequently proposed as an attempt to

resolve security and trust issues in P2P energy trading. However, a massive computational and energetic burden sometimes outweighs the financial viability for small-scale prosumers if the blockchain technology is not properly chosen [25].

Traditional centralized community energy management systems usually adopt a top-down control logic, which optimizes the operation of the community energy system by coordinating generation, storage, and consumption. Centralized coordination considers the overall state of the energy system to achieve high levels of system efficiency. However, these approaches often introduce severe limitations in incentive transparency and user autonomy. Operating as a black box optimization, these centralized systems fail to provide transparent mechanisms for prosumers to jointly negotiate their energy exchanges [21].

The comparison of these coordination mechanisms is summarized in Table 2.2. While these mechanisms provide different approaches to coordinating electricity flows within energy communities, they often face the trade-offs between flexibility, complexity, and transparency. The motivational failure of fixed rules, the regulatory and physical barriers of pure P2P markets, and the black-box nature of centralized energy management system strongly motivate the exploration of alternative mechanisms. These limitations reveal that the challenges of coordination in energy communities go beyond the simple allocation of electricity and involve the governance of contributions, incentives, and shared community resources. Therefore, token-based mechanisms have gained attention due to its potential to act as a socio-technical coordination layer that digitizes governance mechanisms and partially decouple them from physical grid constraints. In this way, token-based mechanisms can represent individual energy contributions while maintaining transparent and flexible governance structures. The concept and design of token-based mechanisms in energy systems will be explained in detail in the following sections.

Table 2.2: Typical coordination mechanisms in energy communities

Mechanisms	**Fixed rule-based energy sharing**	**Market-based P2P trading**	**Centralized community energy management**
Key Principles	Electricity is allocated according to predefined rules.	Community members directly trade electricity through local energy markets or blockchain based platforms.	A central energy management system optimizes the generation, storage, and consumption within the community.

Advantages	Simple implementation; transparent allocation; low operational complexity.	Price signals enable economic optimization; decentralized transaction; potential incentives for behavioral change.	High operational efficiency; coordinated scheduling of distributed energy resources.
Limitations	Inflexible allocation; limited ability to reflect dynamic contributions of participants.	Complex market design; regulatory barriers; potential transaction costs.	Limited transparency for participants; weak individual incentives; reliance on centralized control.

2.2 Energy Flow Modelling in Energy Communities

Before it comes to the tokenization, it's essential to understand the underlying physical energy flows within the community since energy tokens are intended to represent energy-related contributions. The generation, consumption, storage, and sharing of renewable electricity determine where the surplus electricity occurs and how community members contribute to the collective energy system. This chapter reviews the important role of physical energy flow representations and the existing modelling approaches for energy community analysis.

2.2.1 Importance of Physical Energy Flow Representation

A fundamental understanding of physical energy flows is essential before implementing any coordination mechanisms, mainly due to the inherent spatio-temporal variance of distributed energy resources such as PV. At the building level, the mismatch between peak solar generation and peak residential demand creates significant challenges for load matching. Therefore, precise representation of these energy flows is essential to identify the exact magnitude and timing of surplus electricity and quantify critical metrics such as self-consumption (the share of PV production consumed directly) and self-sufficiency (the degree to which on-site generation meets the load) [26]. When scaled to community level, energy communities operate as integrated energy systems, where mapping these physical flows supports the collective management of flexible loads, local balance maintenance and grid congestion mitigation [27].

From a governance and coordination perspective, the physical energy system serves as the foundation upon which any incentive mechanism is built. Lowitzsch et al. (2020) conceptualizes energy communities and their underlying technical networks (i.e., renewable energy clusters) as interrelated socio-technical mirrors [28]. A renewable energy cluster is characterized by the physical interconnectivity of different actors, bidirectional energy flows and the utilization of system flexibility. Consequently, the community's governance model must mirror these physical dynamics. Since tokenization aims to represent energy-related contributions, an accurate physical representation of how energy is generated, stored, and shared within the community is the fundamental prerequisite for the design of token-based governance mechanisms. Without a proper physical modelling layer, token issuance and allocation rules carry the risk of disconnecting from the actual operation of the community energy system.

2.2.2 Existing Energy Community Modelling Approaches

Various modelling approaches have been developed to analyze the operation and performance of energy communities. Instead of viewing different modelling methods as mutually exclusive, recent literature increasingly highlights their complementary roles in addressing different aspects of energy community analysis. Existing studies generally adopt two fundamental modelling approaches, namely simulation-based models and optimization-based models. Each type generate different outputs and offer distinct advantages and limitations.

2.2.2.1 Simulation-based Approaches

Simulation-based models are descriptive and exploratory approaches. They evaluate how energy systems behave over time under predefined behavioral rules and operational logic, essentially answering what happens if a specific mechanism is applied. These models generally produce high granular temporal data which describes electricity generation, consumption, storage operation, energy sharing, and grid interactions within energy communities. Typical outputs include dynamic energy flows, real-time battery state-of-charge (SoC) measurements, dynamic load matching metrics, and the precise magnitude of surplus generation or deficit [29], [30].

A major advantage of simulation-based approaches lies in their ability to capture the heterogeneity of community members and modelling dynamic behavioral changes [31]. By representing energy systems at hourly or sub-hourly resolution, simulation models provide detailed insights into how different system components interact under different operating conditions [32]. They are suitable for evaluating alternative operational rules, coordination mechanisms, and governance structures without constraining the system to a predefined mathematical optimum.

Accordingly, the primary limitation of simulation is its inability to identify optimal system configurations or operating strategies. They evaluate the performance of predefined scenarios, which means the results are dependent on the assumptions and operational rules embedded within the simulation framework [32].

2.2.2.2 Optimization-based Approaches

Conversely, optimization-based models are prescriptive analytical tools designed to identify optimal system configurations or operational strategies under a defined set of mathematical constraints and objective functions. In energy community studies, linear programming (LP), mixed-integer linear programming (MILP), and multi-objective optimization models are commonly used to support infrastructure planning and operational decision-making. Typical outputs include optimal sizing of community assets such as PV and BESS, cost-optimal dispatch schedules, self-sufficiency levels, and economic indicators such as net present value (NPV) and total system costs [29].

The core advantage of optimization-based approaches is their ability to establish the techno-economic boundaries of an energy community and support the investment planning decisions [30]. Through the systematical exploration of solution spaces, these models are useful to identify technology configurations and operational strategies that optimally satisfy the predefined objectives [32].

Despite their strength in system planning, optimization models inherently assume future conditions, system operation, decision-making behavior, and completely rational economic actors [30]. As a result, they may provide limited insights into the dynamic operational interactions that emerge during real-world community operation.

2.2.2.3 Implications for Token-Based Governance Design

It’s essential to understand the capabilities of these modelling approaches to develop effective coordination mechanisms within energy communities. Although simulation-based and optimization-based approaches are used for different purposes, they provide complementary insights to support the design of coordination systems in energy communities.

Optimization models can identify suitable technology configurations and system sizes, thereby establishing the physical boundary conditions for the operation of governance mechanisms. In contrast, simulation models can provide detailed representations of system behavior over time, which captures the interactions between electricity generation, consumption, storage operation, and energy sharing under realistic operating conditions.

Recent studies increasingly suggest a combination of both approaches within integrated modelling framework [30]. These integrated approaches provide a more comprehensive understanding of energy community performance by linking long-term system design with short-term operational dynamics. From a governance perspective, this integration is meaningful since the effectiveness of any governance mechanism depends not only on its internal rules but also on the physical conditions under which it is applied. By combining optimization and simulation approaches, governance strategies can be evaluated within physically realistic and techno-economically grounded energy community configurations.

2.3 Tokenization in Energy Systems

Tokenization refers to the process to convert assets, rights, or contributions into digital units that can be recorded, transferred, and managed within a digital system [33]. These digital units are commonly referred to as tokens, which act as a representative layer that links physical activities or resources to digital record of value. Building on the representation of physical energy flows discussed in the previous chapter, tokenization in energy systems provides a potential mechanism to translate physical energy flows into digital representations that can be tracked and allocated among participants.

2.3.1 Complementary Currencies and Digital Tokens

To understand how token-based governance mechanisms function in energy systems, it is essential to bridge the gap between the economic concept of complementary currencies and the technological implementation of digital tokens.

Complementary currencies (CCs) are defined as alternative media of exchange designed to operate alongside rather than replace the official national fiat money [34]. CCs have been designed to connect underutilized local resources with unmet community needs. Different from conventional currency, which is primarily driven by macro-economic accumulation, a community-oriented currency can be optimized to encourage local circulation, strengthen social capital and incentive sustainable behaviors. In the context of energy communities, CCs can therefore be applied to support the coordination between energy generation and demand and guide the behavior change of community members.

Digital tokens have emerged as a flexible tool to represent value, rights, or contributions within digital systems. In the field of distributed ledgers, digital tokens represent a technological evolution of these alternative currencies. Depending on the functional design, digital tokens in energy sector are generally classified into three archetypes: payment tokens for local settlement, utility tokens that grant specific

access rights to energy utilities or services, and security tokens that represent a digital claim on energy assets [35].

From a convergence perspective, digital tokens can be interpreted as a technological implementation of complementary currency principles in digital systems. A key feature which differentiates digital tokens from traditional complementary currencies is their programmability. The issuance and circulation of tokens can be autonomously synchronized with physical energy events through digitally defined governance rules [24]. For example, token issuance can be algorithmically tied to the exact moment when a smart meter detects the injection of renewable energy, while its value can be programmed to decay over time or expire at a certain time to discourage hoarding and encourage immediate local consumption [35]. Therefore, digital tokens provide an essential technical architecture to achieve the socio-economic goals of complementary currencies while coordinating complex interactions within energy communities.

2.3.2 Tokenization of Energy

The tokenization of energy involves abstracting physical energy flows into programmable digital representations that can be recorded and managed within digital systems [36]. In modern energy systems, the booming installation of digital monitoring infrastructures, such as smart meters and distributed energy management platforms, enables the accurate measurement of electricity generation, consumption, and storage. These measurable energy activities set the necessary foundation for translating physical flows into digital representations.

By mapping a specific amount of energy (e.g., 1 kWh injected into grid) or a quantified reduction in consumption to a digital token, the system creates a link between the physical distribution grid and a virtual coordination layer [22]. These digital tokens can therefore function as records, reflecting individual participation in the energy community.

Based on current research and pilot projects, several applications of energy tokenization have been explored in recent years [36]. One example is the use of tokens as incentives to accelerate the low-carbon transition without relying solely on state-funded subsidies. By algorithmically issuing tokens proportional to the amount of generated renewable energy, a continuous incentive structure to reward generation can be created. For demand-side management, energy savings can be tokenized by comparing actual consumption against a business-as-usual baseline to reward users for energy conversation efforts. Other applications include token-based settlement mechanisms for local energy trading [22] and digital certificates for tracking renewable energy attributes [23]. To summarize, tokenization of energy create opportunities to design various incentive mechanisms for the governance and coordination of distributed energy resources within energy communities.

2.3.3 Tokens as Governance Mechanisms

Beyond serving as digital representations of energy-related activities, tokens can further act as governance mechanisms for coordinating contributions, incentives, and access to shared resources within energy communities. Because tokens are programmable, predefined governance rules can be embedded into token issuance, allocation, and circulation processes. This translates community objectives into operational mechanisms that influence how contributions are recognized, how benefits are distributed, and how shared resources are utilized. Therefore, more granular and transparent coordination of collective resources is enabled through two main functions, namely behavior steering and resource allocation.

Behavior steering is an important governance function of tokens. To maximize the utilization of local renewable energy resources, energy communities require continuous coordination between electricity generation, consumption, and storage. Rather than relying solely on centralized control mechanisms, tokenized governance utilizes principles of motivational psychology and behavioral economics [22]. Incentive structures can be embedded in token-based systems to encourage community members to voluntarily adopt behaviors aligned with collective objectives. For instance, the value or issuance rate of a token can be designed to fluctuate dynamically based on grid constraints, either rewarding users who shift their loads to periods of peak solar generation or penalizing those who exceed consumption baselines during grid stress [36]. In this way, tokens become behavioral signals that connect individual decisions with community goals and further help align private incentives with collective benefits.

Another important governance function of tokens is related to the allocation of shared resources and collective benefits. Examples of common resources within energy communities include shared battery storage, locally generated renewable electricity, or revenues derived from grid exports. While traditional allocation rules based on fixed ownership shares may fail to reflect actual participation, token-based mechanisms enable more granular contribution-based allocation. By linking community benefits or resource access to measurable forms of participation, such as renewable energy generation, surplus electricity utilization, or engagement in community activities, token systems can record individual contributions more transparently [20]. This helps to mitigate the free-rider problem and ensure a demonstrably fair distribution of shared resources.

Ultimately, token-based governance provides a mechanism for energy communities to coordinate decentralized actions while maintaining transparency and flexibility. It encodes the values of the community, whether they prioritize economic efficiency, poverty alleviation, or maximum carbon reduction, directly into the mathematics of the token’s issuance, allocation, and circulation rules [23]. In other words, tokens help

communities define how contributions are recognized, how benefits are distributed, and how collective objectives are pursued. In this sense, tokenization extends beyond a technical accounting system and becomes a governance framework which empowers citizens to jointly manage shared energy resources more transparently and equitably than traditional hierarchical approaches to energy management.

2.4 Comparative Analysis of Energy Token Projects

A growing number of experimental projects have explored the application of energy tokens in recent years. These incentives have significant difference in their objective, technological implementation and governance design. Some aim at incentivizing local renewable energy generation, while others focus on facilitating local energy trading or strengthening community participation. Together, these projects have shown how token-based mechanisms can be integrated into energy systems to represent energy-related activities and coordinate stakeholders. By reviewing these representative energy token projects and comparing their design logic across multiple dimensions, this section aims to identify common design patterns, extract key principles for token-based coordination mechanisms, and further derive a conceptual framework for energy token design.

2.4.1 Overview of Energy Token Projects

In total, seven energy token projects have been selected to explore the integration of digital tokens into energy systems. Rather than viewing them as isolated case studies, this section organizes them following a conceptual continuum, which evolves from global generation focused incentives to localized, behavior-driven, and socio-economically embedded systems.

SolarCoin [37] presents the most macroscopic approach, which is a global reward program for solar PV installation. Its primary objective is to provide an additional symbolic incentive for renewable energy deployment. Based on a static peg, one SolarCoin is issued for every 1 MWh of verified solar electricity produced. Even though scalable, this mechanism is largely grid blind. It does not account for local grid constraints or temporal demand matching. Instead, it functions as a recognition mechanism rather than a dynamic energy system management tool.

To address the spatial and temporal limitations, NRGcoin [38] introduces a local grid balancing logic. Designed for decentralized electricity networks, NRGcoin dynamically mints tokens only for injected green energy that is consumed locally within a specific 15-minute time slot. This temporal alignment actively discourages over-production and mitigates grid stress by rewarding prosumers based on real-time local demand.

Shifting the focus from energy generation to consumption, ATOMcoin [36] is proposed as a dynamic behavioral steering tool. Its issuance is tied to “negawatts”, which means the amount of energy saved relative to a forecasted business-as-usual baseline. Through algorithmic pricing adjustments and penalties for excess consumption, ATOMcoin encourages prosumers to change their behaviors towards peak-load shaving and more efficient usage patterns.

Ergo [39] explores an approach that aligns token circulation with the physical constraints of renewable energy availability. The issuance of Ergo is conceptually linked to the total forecasted renewable supply over a defined time horizon. Mechanisms such as limited validity periods or demurrage are incorporated to discourage long-term accumulation and encourage the utilization of energy services within a specific timeframe. Therefore, Ergo has the potential to reflect the operational characteristics of renewable energy systems.

A further group of projects focus on how energy tokens can address local socio-economic challenges rather than being limited by purely technical energy coordination. The Vilawatt project [40] aims to incentivize energy efficiency and renewable adoption in the Spanish city of Viladecans through a local digital token, as well as redirect purchasing power toward local businesses. Households accumulate energy points based on energy savings, solar PV installation, participation in sustainability programs. The issuance is based on verified sustainability performance by the municipality. The Stanz Token [35] in Austria explores the merging of a renewable energy community utility token with a municipal voucher system. This hybrid structure combines energy incentives with local economic instruments, which helps to enhance rural autonomy and strengthen community resilience. Similarly, PalmaSolar [41] in Brazil tackles energy poverty in low-income areas. Families purchase renewable energy from the community solar plant through a local community bank at lower prices without upfront capital. 50% of their energy payments return to them as cashback in the form of a local solar currency which can be used for local trade. In parallel, part of the funds support micro-loans for local entrepreneurs, thereby linking renewable energy projects with broader community development goals.

These case studies demonstrate that token-based mechanisms can play multiple governance roles in energy systems, varying from renewable energy incentives and behavioral steering to community coordination and socio-economic development. The diversity of these implementations calls for a flexible analytical framework which can extract different token design ideas and the corresponding governance strategies.

2.4.2 Comparative Analysis

Several common design dimensions can be summarized to compare these energy token projects. Table 2.3 presents the comparative analysis based on four aspects,

namely value anchor, issuance logic, circulation mechanism and governance role. Through this comparison, we can have a better understanding of how tokens represent energy-related value, how they are generated, how they circulate within the system, and how they influence participant behavior.

Regarding the value anchor, early initiatives like SolarCoin simply link tokens to renewable electricity generation. More recent projects expand this concept by anchoring token value to broader energy system contributions, such as local energy balancing, efficiency enhancement, or participation in community activities. This shift indicates a gradual transition towards more complex coordination-oriented token design.

There are also differences in the issuance logic of tokens. Some projects use relatively simple issuance rules, where tokens are minted proportionally to renewable electricity generation. Others integrate more complex incentive structures into issuance, such as temporal balancing conditions, behavioral incentives, and forecasted renewable supply constraints. These variations imply that issuance rules can be flexibly designed to encode different incentive structures in the energy system.

Circulation mechanisms further differentiate the design of tokens. Some tokens are utilized for open digital circulation, while others are designed with restrictions to localized ecosystems, where they can be exchanged for community services, energy-bound benefits, or local economic activities. These differences influence not only how tokens circulate, but also the governance functions they perform within the communities. As a result, the governance role of tokens varies across projects, ranging from simple renewable energy incentives to community coordination and broader socio-economic engagement.

Table 2.3: Comparative analysis of selected energy token projects

Project	Value Anchor	Issuance Logic	Circulation Mechanism	Governance Role
SolarCoin	Solar electricity generation	1 token per MWh solar generation	Open blockchain circulation	Renewable generation incentive
NRGcoin	Renewable electricity injected and locally consumed	Tokens issued for locally balanced renewable energy within time intervals	Exchangeable within local markets	Encourage local energy balancing and grid stress mitigation

ATOMcoin	Energy savings (Negawatts)	Tokens issued based on energy efficiency improvements	Platform-based circulation	Behavioral steering toward energy efficiency
Ergo	Renewable energy availability	Token issuance linked to expected renewable supply	Limited validity period (demurrage)	Align energy consumption with physical supply
Vilawatt	Energy savings and community participation	Tokens issued for efficiency improvements	Circulation within municipal ecosystem	Strengthen local circular economy
Stanz Token	Renewable energy community participation	Tokens issued for participation in REC activities	Circulation within local voucher system	Enhance rural community governance
PalmaSolar	Community solar energy participation	Tokens issued through community energy program	Circulation within local social economy	Address energy poverty and support local development

2.4.3 Analytical Framework for Energy Token Design

Based on the comparative analysis, several recurring design dimensions can be identified across existing energy token projects. Although these projects pursue different objectives, their token mechanisms can generally be described through four key dimensions, namely value anchor, issuance trigger, circulation mechanism, and supply control. An analytical framework is therefore developed for the design of energy tokens, shown as Figure 2.2. This framework links the characteristics of the target energy system with the design of token mechanisms.

Given a certain energy community, the first step is to identify the system objective which defines the coordination problem addressed by the token system. Then, the token system can be designed through four key design dimensions. The value anchor specifies the physical activity or contribution represented by the token within an energy community. The issuance trigger sets the conditions for the creation of new tokens to translate energy-related events into digital token units. The circulation mechanism

defines how tokens flow in the system and how participants can use them. The supply control regulates the availability of tokens, through fixed rules, algorithmic minting, time-based mechanisms and so on.

Generally, this framework can be used as a conceptual foundation for the development of token-based governance methodology in this thesis.

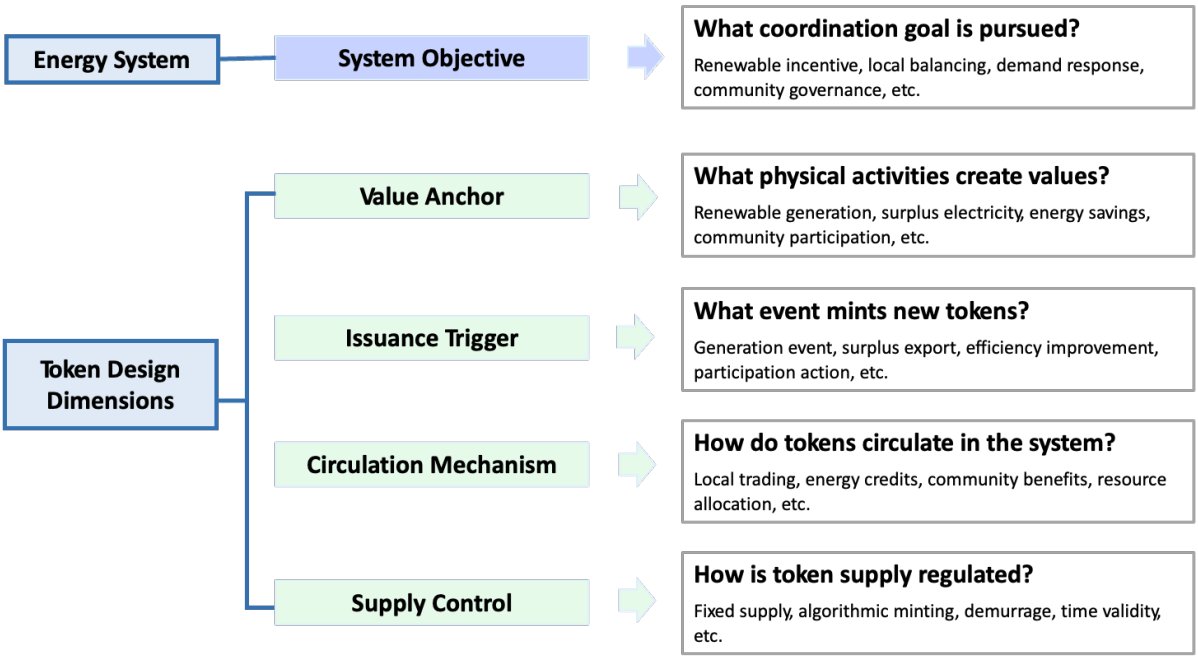


Figure 2.2: Analytical framework for energy token design

2.5 Regulatory Foundations for Token-Based Energy Communities

While token-based mechanisms provide new approaches to coordinate energy-related activities within energy communities, the design of energy tokens is not merely determined by technical feasibility and governance objectives. The regulatory environment in which the energy community operates should also be taken into consideration. Depending on the design, tokens may interact with energy market regulations, financial legislation, consumer protection requirements, and data protection rules. As a result, regulatory considerations play a vital role in determining the feasible design space of token-based governance mechanisms.

This chapter reviews the regulatory foundations related to energy communities and energy tokenization from both the EU and Swedish perspectives. Different token classifications and the associated regulatory implications are further discussed. Based on this, a regulatory design boundary is finally derived to identify the opportunities, constraints and feasible design space for token design in energy communities.

2.5.1 EU Framework

As introduced in Section 2.1, the European Union has established the legal foundation for energy communities through the Clean Energy for All Europeans Package, especially through the introduction of RECs and CECs. Building upon this foundation, this section focuses more on the regulatory rights and activities enabled by the EU framework and the implications for the design of token-based governance systems.

From a token design perspective, the most relevant aspect of the EU framework lies in the range of activities that the energy communities are permitted to undertake. The current regulations allow energy communities to collectively generate, consume, store, and share energy, as well as sell electricity [14], [15]. Furthermore, activities such as demand response, aggregation, energy efficiency services, and the use and management of shared energy resources are supported within the broader framework for citizen participation in energy systems.

These activities represent measurable forms of participation that have the potential to be recognized and incentivized through token mechanisms. The potential value anchors for token-based governance systems may include renewable energy generation, surplus electricity utilization, participation in flexibility services, and contributions to community energy management. Consequently, the EU framework provides a legal basis for many of the energy-related activities that token systems may seek to coordinate. Meanwhile, the existing EU framework has no explicit regulations for energy tokenization or prescribe how such coordination mechanisms should be implemented. This means the design of token-based governance mechanisms remains largely an open area for innovation within the broader regulatory boundaries for energy communities.

2.5.2 Swedish Context

Even though the EU has established a common framework for energy communities, the actual implementations depend largely on national legislation. In Sweden, the transposition of the Clean Energy Package remains an ongoing process. Compared with several other European countries, Sweden has not yet introduced a dedicated regulatory framework specifically for energy communities. Community-based energy sharing arrangements are rarely mentioned since the existing support measures are primarily focused on renewable energy deployment and energy efficiency improvements [18]. Therefore, the practical implementation of energy communities still relies mainly on the existing electricity market structure.

A key characteristic of the Swedish electricity system is its continuous reliance on regulated distribution networks operated by distribution system operators (DSOs). As a result, local energy exchanges within energy communities remain subject to the

broader regulatory and market arrangements which govern the public electricity grid. Previous studies have highlighted that the attractiveness of market-based P2P trading models may be reduced due to limited economic incentives for local energy sharing and the absence of dedicated mechanisms for coordinating community-level energy exchanges under the Swedish context [18].

From the perspective of token-based governance, these conditions suggest that regulatory feasibility should consider the compatibility of technical design of the token system with existing market structures. Therefore, it's essential to understand the Swedish regulatory context to identify which forms of token-based coordination can realistically be implemented within energy communities.

2.5.3 Token Classification and Regulatory Implications

Besides its technical design, the regulatory implications of a token system depend also on how a token is functionally classified. Different token designs may serve different objectives within an energy community, such as accessing services, incentivizing sustainable behaviors, and facilitating the settlement of transactions. Therefore, different token types may have different interactions with the regulatory framework.

Building upon the discussion of complementary currencies and digital tokens in Section 2.3, this section focuses on the classification of tokens from a functional and regulatory perspective. It should be noted that token classifications can be approached from different perspectives. Section 2.3 introduced common token categories discussed in digital asset literatures, but the focus here is on the functional roles that tokens may play within energy communities. Therefore, three categories relevant to energy communities are summarized, namely utility tokens, complementary currencies, and settlement tokens. Understanding the characteristics of these token types is essential for assessing their suitability and regulatory implications for token-based energy community governance systems.

2.5.3.1 Utility Tokens

Utility tokens refer to digital tokens that provide access to specific services, functions, or rights within a defined ecosystem. According to the regulatory classifications commonly adopted in digital asset literatures, the primary purpose of utility tokens is usually linked to the utilization of a certain service, participation in a platform, or access to a specific function within a system [42], [43]. They are designed to grant users access rights rather than ownership claims or monetary value.

In energy communities, utility tokens can be applied to facilitate access to shared energy resources, community-related services, participation in community affairs, and so on [35]. Since the value of a utility token is primarily derived from its functionality

rather than the exchange value, approaches based on utility tokens are generally aligned with the participatory and service-oriented nature of energy communities [44]. As a result, utility tokens are often discussed as potential instruments for governance within decentralized energy systems.

2.5.3.2 Complementary Currencies

As explained in Section 2.3, complementary currencies are alternative exchange mechanisms designed to operate alongside national fiat currencies. Their primary purpose is to encourage specific economic, social, or environmental objectives within a defined community [45]. By nature, complementary currencies have the potential to promote local value circulation and incentive behaviors which contribute to the collective goals.

For energy communities, mechanisms based on complementary currencies can encourage behaviors that support local sustainability objectives by linking rewards to activities such as renewable energy generation, energy conservation, flexibility contribution, and community engagement [45], [46]. Therefore, complementary currencies are usually proposed as instruments for promoting community participation and supporting the common sustainability goals within energy communities.

2.5.3.3 Settlement Tokens

Settlement tokens are digital tokens designed for facilitating the exchange and settlement of transactions between participants. Different from utility tokens and complementary currencies mentioned above, settlement tokens function as a medium for transferring value within a defined system. The value of settlement tokens is often connected with a specific asset, monetary unit, or exchange mechanism which enables transactions between users [43].

The most common scenario in which settlement tokens are applied in energy communities is facilitating local energy trading and P2P electricity transactions. Under this scenario, tokens can be used to represent energy-related value and support the settlement of exchanges between market participants [25]. This means settlement tokens are more closely associated with payment and transaction functions rather than community governance. From the perspective of regulation, settlement-oriented token designs may need additional considerations since they operate closer to monetary exchange mechanisms.

2.5.3.4 Regulatory Implications

The classification of a token helps to explore its potential interaction within existing regulatory frameworks. Although the boundaries between different token categories

are not always explicit, the function emphasis of utility tokens, complementary currencies, and settlement tokens normally differ within a system. Utility tokens focus more on access to services and participation rights, complementary currencies emphasize behavioral incentives and community objectives, while settlement tokens are more used for value transfer and transaction settlement.

In terms of regulation, existing frameworks such as the Markets in Crypto-Assets Regulation (MiCA) distinguish between tokens providing access to goods or services and tokens performing payment-related functions [43]. As a result, token systems designed primarily for transaction settlement may require extra regulatory consideration in comparison with governance-oriented mechanisms. Utility-token and complementary-currency approaches may provide greater regulatory compatibility than settlement token designs in energy communities, where token systems are primarily intended to encourage stakeholder participation, coordinate collective actions, and support sustainability objectives.

2.5.4 Regulatory Design Boundary

Although existing regulations do not explicitly prescribe the rules for the design of energy tokens, they establish several conditions which influence the feasible design space. Therefore, it's useful to translate these regulatory observations from the above sections into practical design implications for token development.

Table 2.4 summarizes the regulatory design boundary derived from the EU framework for energy communities, the Swedish context, and the functional classification of energy tokens. These implications provide guidance to identify the proper token design approaches, which are more compatible with the current regulatory environment.

Table 2.4: Regulatory design boundary for energy community tokens

Regulatory Observation	**Token Design Implication**
RECs and CECs are legally defined as entities with primary purpose to provide environmental, economic, or social benefits rather than financial profits.	Rather than pure profit maximization, token mechanisms should support community-oriented objectives and collective benefits.
Energy communities are allowed by the EU framework to engage in renewable energy generation, self-consumption, storage, energy sharing, flexibility	Token value anchors and reward mechanisms should be linked to energy-community activities which are legally recognized.

provision, and other community energy activities.	
The Swedish energy system relies on the regulated public grid infrastructure and existing electricity market arrangements.	Tokens should function as a virtual coordination layer instead of ownership representation or physical transfer of electricity.
There is currently no dedicated legal framework for energy tokenization in energy communities based on existing regulations.	Token design should remain transparent, simple, and compatible with existing energy-community structures.
Utility tokens and complementary-currency mechanisms are primarily linked to access, participation, and behavioral incentives.	Governance, participation, and incentive functions should be prioritized within the token system.
Settlement tokens operate closer to payment and value-transfer functions and may introduce additional regulatory considerations.	It should be avoided to involve direct payment, monetary settlement, and speculative functions.

The derived regulatory design boundary suggests that energy tokens should serve as governance mechanisms associated with legally recognized activities within energy communities. Token design should focus more on supporting community participation and coordination to realize collective sustainability objectives. Moreover, due to regulatory risk, direct payment and settlement functions should be largely avoided to distinguish token-based governance mechanisms from monetary exchange mechanisms.

2.6 Research Gaps and Analytical Implications

The literature reviewed in this chapter positions energy-token design as a socio-technical governance problem rather than solely a technical or transactional one. Physical energy-system models provide methods for representing renewable generation, electricity demand, storage operation, and grid interaction, while existing token initiatives demonstrate diverse approaches to value representation, issuance, circulation, behavioral incentives, and community participation. The feasible functions of these mechanisms are further shaped by the regulatory environment in which an energy community operates. Despite the relevance of each research area, their implications for the integrated design and evaluation of energy-token governance

remain only partially connected. The following synthesis therefore identifies four areas that require further consideration.

The first research gap concerns that relationship between physical energy-system operation and token issuance. Energy-community models can quantify renewable generation, surplus electricity, storage operation, and grid exchange. At the same time, the token projects reviewed in this chapter apply different value anchors and issuance rules, including renewable-energy generation, energy savings, local balancing, and participation in community activities. However, these research streams do not establish a common approach for determining which physically realized energy contributions should be eligible for token issuance. Addressing this gap requires a clear connection between modelled community energy flows and the conditions under which tokens are created. This analytical requirement informs the physical-layer modelling and token-issuance methodology used to address RQ1.

The second research gap concerns the integration of different token-governance functions. The reviewed projects show that tokens may be used to represent contributions, distribute benefits, encourage behavioral change, provide access to services, support local circulation, and facilitate community participation. However, these functions appear in different combinations and are generally designed around the specific objectives of individual projects. As a result, their interactions and combined implications for the coordination of community resources remain difficult to compare. This creates a need for an integrated governance framework in which alternative allocation, circulation, incentive, and supply-control mechanisms can be represented under consistent physical conditions. This requirement informs the governance-layer analysis conducted under RQ2.

The third research gap relates to the evaluation and comparison of alternative token designs. The comparative analysis identified substantial variation in token value anchors, issuance triggers, circulation mechanisms, and supply-control approaches. These design choices reflect different governance priorities and may produce different outcomes for individual members and the community as a whole. Consequently, the performance of a token mechanism cannot be assessed adequately through a single indicator or inferred from its design features alone. A multidimensional evaluation is therefore needed to assess performance across community activation, distributional fairness, token-enabled economic value, and token persistence, while identifying the trade-offs that emerge among different governance objectives. This analytical implication provides the basis for the comparative design evaluation conducted under RQ3.

The fourth research gap concerns the relationship between technically feasible token functions and their compatibility with the applicable regulatory environment. As discussed in Section 2.5, the EU framework recognized a range of activities undertaken by energy communities, including renewable-energy generation,

consumption, storage, energy sharing, flexibility provision, and energy services. However, the Swedish context does not currently provide a dedicated legal framework for energy tokenization, and token functions involving payment or transaction settlement may require additional regulatory consideration. Regulatory feasibility therefore depends not only on the technical operation of a token mechanism but also on its functional role and the activities it is intended to support. This finding motivates the assessment of regulatory constraints and design boundaries conducted under RQ4.

Taken together, these research gaps support a layered analytical approach to the design and evaluation of energy tokens. The physical layer establishes the community energy flows and identifies the renewable-energy contributions available for tokenization. Based on these physical conditions, the governance layer represents the issuance, allocation, circulation, redemption, and use of tokens as community incentives. Alternative physical and governance configurations are then compared through a multidimensional evaluation framework, while the regulatory analysis is used to interpret their feasible functions and implementation boundaries. This layered structure provides the conceptual basis for the methodology presented in Chapter 3.

3 Methodology

3.1 Overall Framework

Building on the research gaps and analytical implications identified in Chapter 2, the thesis proposes an integrated four-layer framework for renewable energy communities that establishes a direct connection between physical energy operation and token-based community governance, as illustrated in Figure 3.1. This framework combines renewable energy system design and operation, energy tokenization, behavioral coordination, and value realization into a unified architecture. Instead of treating energy tokens solely as economic incentives or digital assets, the framework anchors token creation in measurable renewable energy contributions and connects token-based incentives with demand-side coordination through a structured physical-governance interface.

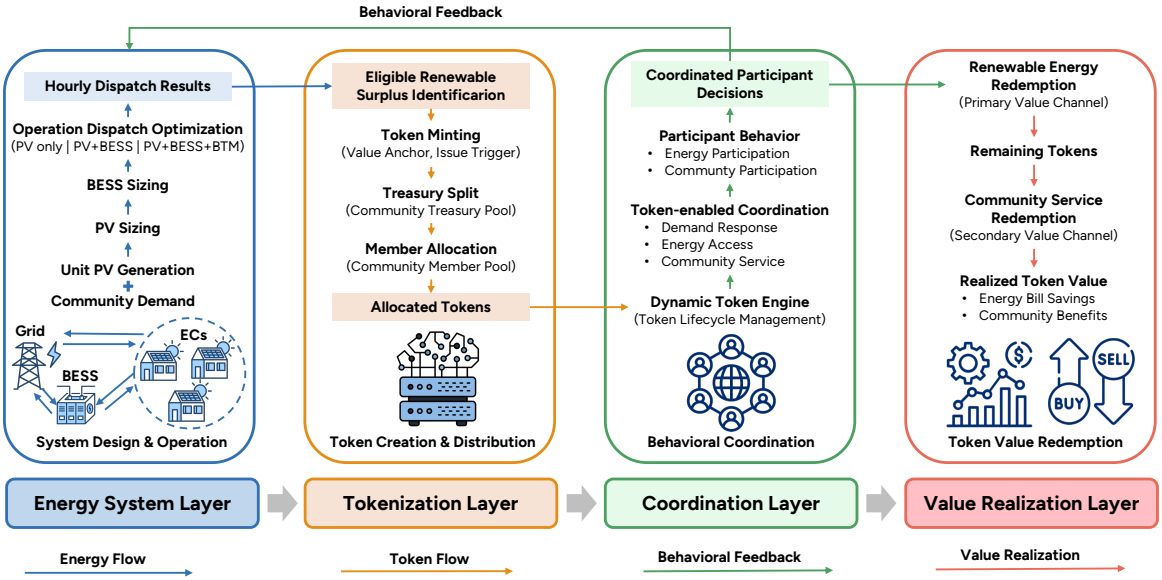


Figure 3.1: Integrated Physical-Governance Framework

The Energy System Layer sets up the physical foundation by designing and operating the community energy system through PV sizing, battery sizing, and operational dispatch optimization. It produces hourly dispatch results that describe the time-resolved distribution of renewable energy within the community. Based on these physical outcomes, the Tokenization Layer identifies eligible renewable energy surplus as the value anchor for token issuance and converts renewable energy contributions into digital tokens through token minting and distribution mechanisms. The generated tokens are subsequently managed within the Coordination Layer, where dynamic token management and token-enabled coordination mechanisms influence participant decisions regarding energy participation and community participation. Finally, the

Value Realization Layer enables token holders to redeem accumulated tokens, prioritizing renewable energy redemption before allowing remaining tokens to be exchanged for community services. Through these redemption channels, digital tokens are transformed into tangible community benefits, including reduced electricity expenditures and access to community services.

Four complementary interactions connect these layers. The Energy Flow represents the physical progression from renewable energy generation to optimized community dispatch. The Token Flow transforms eligible renewable surplus into community-owned digital tokens. Behavioral Feedback represents the governance-layer pathway through which token-based incentives are translated into simulated demand-side responses. In this study, these responses are represented through rule-based demand-response and token-utilization mechanisms rather than through endogenous modelling of individual participant behaviors. Finally, Value Realization describes the process through which redeemed tokens generate measurable economic and social value within the community.

Different from the conventional token-based community models, which typically terminate at token issuance or financial reward, the proposed framework extends the governance process to token allocation, circulation, demand-side coordination, and value realization. By connecting physically grounded token issuance with rule-based behavioral coordination, the framework enables alternative governance mechanisms to be evaluated within a common community energy context. This structured connection also provides a foundation for future bidirectional integration, in which governance responses and physical system operation could be coordinated more dynamically.

3.2 Case Study

The proposed physical-governance framework was evaluated through a residential renewable energy community located in Tamarinden, Örebro, Sweden. This section first introduces the characteristics of the case study, followed by the input data adopted for the simulation and the principal assumptions used throughout the analysis.

3.2.1 Tamarinden Energy Community

Tamarinden is a newly developed residential district in Örebro, established as one of the country's pioneering neighborhoods for sustainable urban development and local energy collaboration [47]. The district has been planned with the objective of integrating renewable electricity generation, energy-efficient buildings, and coordinated energy management to reduce dependence on the external electricity grid while promoting local utilization of renewable energy. Instead of treating individual

buildings as isolated energy consumers, Tamarinden has been designed to facilitate cooperation among buildings, making it a representative case for evaluating renewable energy communities.

The Tamarinden energy community comprises ten residential apartment buildings, which are denoted as Buildings A-J for modelling purposes through this study, as illustrated in Figure 3.2 [47]. These buildings are modelled as a single renewable energy community, where electricity demand, renewable energy generation, battery energy storage, and energy exchanges are evaluated at the community level. This community forms the basis for both the physical energy system analysis and the subsequent governance-layer evaluation.



Figure 3.2: Overview of the Tamarinden residential district in Örebro

3.2.2 Input Data

The proposed framework integrates multiple datasets to support the physical energy simulation and governance-layer evaluation. These datasets include community electricity demand, photovoltaic (PV) generation, meteorological data, and electricity prices. All datasets were processed at an hourly temporal resolution for the year 2019 to ensure temporal consistency throughout the simulation framework. The year 2019 was selected because complete and temporallly consistent datasets were available for all required input variables.

3.2.2.1 Community Electricity Demand

Hourly electricity demand data consist of hourly demand profiles for the residential buildings A-J within the Tamarinden energy community. For the community-level

physical energy simulation, the building-level demand profiles were aggregated to get the total hourly electricity demand. Meanwhile, the original building-level data were retained to represent the heterogeneity of electricity consumption among community members in the subsequent governance-layer analysis.

3.2.2.2 Photovoltaic Generation

The original PV generation dataset consists of hourly generation profiles for the ten residential buildings A-J. These building-level profiles were used to represent differences in renewable electricity contribution among individual community members in the governance-layer analysis, particularly for the member-level token allocation. The PV generation used in the physical-layer simulation was generated separately by REUnity tool based on the selected PV system capacity and location-specific meteorological inputs.

3.2.2.3 Meteorological Data

Meteorological data required for the PV generation simulation were obtained using the geographical coordinates of the Tamarinden community through the ERA5 and PVGIS APIs. PVGIS [48] was used to retrieve hourly solar irradiance, ambient temperature, and other meteorological variables required for PV performance simulation. Hourly snowfall and snow depth profiles were obtained from the ERA5 dataset via the Copernicus Climate Data Store API [49] to support the snow-loss component of the PV simulation model.

3.2.2.4 Electricity Prices

Hourly day-ahead electricity price data were obtained from the ENTSO-E Transparency Platform [50] for the SE3 bidding zone, corresponding to the location of the Tamarinden energy community. In the simulation framework, electricity imported from the external grid was priced using hourly retail electricity rates derived from the day-ahead market prices. Electricity exported to the external grid was remunerated using a spot-price-based feed-in tariff. Electricity exchanged within the community was settled using a mid-market pricing mechanism, where the internal settlement price was defined as the midpoint between the external grid import and export prices. This pricing configuration was applied consistently throughout the physical-layer and governance-layer simulations.

3.3 Physical Layer Modelling

The physical layer models the operation of the renewable energy community by simulating electricity generation, storage, consumption, and grid interaction. Based on

the input datasets described in Section 3.2, it determines physically consistent hourly energy flows that provide the foundation for the subsequent techno-economic assessment and governance-layer token mechanism.

The physical layer modelling is implemented within the proposed REUnity framework, with its overall workflow illustrated in Figure 3.3. The framework is organized into three sequential layers. The design layer determines the system configuration through PV generation modelling and system sizing of both PV and BESS. The operational layer simulates hourly community energy flows under different electricity pricing scenarios. The performance assessment layer evaluates battery degradation and lifecycle techno-economic performance. The individual modelling components are described in the following sections 3.3.1-3.3.7.

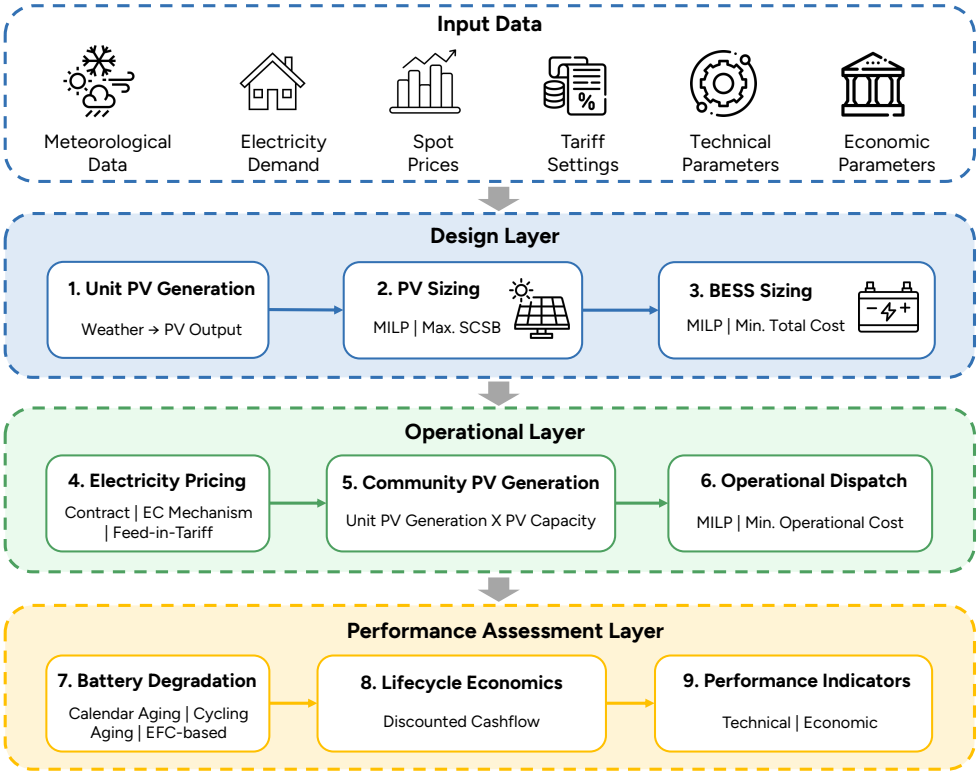


Figure 3.3: Physical layer modelling workflow of REUnity framework

3.3.1 Community Energy Flow Model

The renewable energy community is represented as an interconnected system consisting of multiple participating buildings. Each building is characterized by its own hourly electricity demand and rooftop photovoltaic (PV) generation profile, while the buildings are coordinated at the community level for energy sharing and system operation. Depending on the selected physical configuration, a community-level

battery energy storage system (BESS) may be integrated to store surplus PV electricity and support subsequent community demand. All participating buildings remain connected to the public distribution grid. The model is developed at an hourly temporal resolution, allowing electricity generation, consumption, storage operation, energy sharing, and grid interaction to be simulated consistently throughout the analysis period.

Within each time step, locally generated PV electricity is first used to satisfy on-site demand. Surplus electricity can be stored in the BESS, shared within the community, or exported to the utility grid, depending on the physical configuration and system state. When local generation is insufficient, electricity demand is supplied through battery discharge when available, with the remaining deficit imported from the external grid. This energy flow representation forms the physical basis for the subsequent system sizing, operational dispatch, and performance assessment models.

3.3.2 Unit PV Generation Model

The unit PV generation model converts site-specific meteorological observations into standardized hourly photovoltaic (PV) generation profiles, which provides the renewable energy input for the subsequent PV sizing, operational dispatch, battery degradation, and lifecycle economic assessment models.

The PV generation model consists of three sequential computational stages. First, hourly plane-of-array (POA) irradiance obtained from meteorological datasets is converted into a normalized unit-generation profile. A constant system loss factor is applied to account for inverter, wiring, availability, and soiling losses [51]. Second, temperature-dependent performance variations are incorporated using a NOCT-based cell temperature model [52], where the estimated cell temperature is converted into a temperature correction factor according to the module temperature coefficient. Third, a dynamic snow model computes an hourly snow correction factor representing the combined effects of snow accumulation, melting, and sliding on the PV surface [53]. The snow correction factor is applied multiplicatively to the PV generation profile. The resulting unit-generation profile represents the hourly electricity production per installed kilowatt-peak (kWp) of PV capacity and forms the basis for the subsequent system sizing and operational simulation.

The hourly unit PV generation is calculated as:

$$P_t^{PV,unit} = \frac{G_t}{1000} f^{loss} f_t^{temp} f_t^{snow} \tag{3-1}$$

where G_t is the plane-of-array irradiance, f^{loss} is the constant system loss factor, f_t^{temp} is the temperature correction factor, and f_t^{snow} is the snow correction factor.

In addition to estimating hourly PV generation, the model evaluates the maximum technically installable PV capacity from rooftop characteristics. The usable rooftop area is estimated as:

$$A_{usable} = \eta_{util} A_{roof} \tag{3-2}$$

where η_{util} is the rooftop utilization factor and A_{roof} is the total available rooftop area.

The corresponding maximum technically installable PV capacity is then calculated as:

$$P_{PV}^{max} = A_{usable} \rho_{PV} \tag{3-3}$$

where ρ_{PV} is the nominal module power density. The resulting technical potential defines the upper bound of the feasible PV design space and serves as the capacity constraint for the PV sizing optimization described in Section 3.3.3.

3.3.3 PV Sizing

The PV sizing model determines the installed PV capacity within the technically feasible capacity range defined in Section 3.3.2. The objective is to identify a PV capacity that maximizes renewable energy utilization within the energy community while respecting the maximum rooftop installation potential. The resulting installed PV capacity is used to generate hourly PV generation profile, which serves as the renewable energy input for the subsequent battery sizing and operational dispatch models.

In this study, PV sizing is formulated as a single-objective optimization problem that maximizes the Self-Consumption-Self-Sufficiency Balance (SCSB) [54]. The SCSB index evaluates the balance between local renewable electricity utilization and community energy autonomy by combining the self-consumption rate and self-sufficiency rate into a single performance indicator. Compared with optimizing either indicator individually, the SCSB promotes balanced improvements in both renewable electricity utilization and community energy autonomy.

The installed PV capacity is constrained by the maximum technical installation potential:

$$0 \leq P_{PV} \leq P_{PV}^{max} \tag{3-4}$$

where P_{PV} is the installed PV capacity and P_{PV}^{max} is the maximum technically installable PV capacity determined from rooftop characteristics.

For a given installed PV capacity, the normalized PV generation profile $P_t^{PV,unit}$ is converted into the community PV generation profile as:

$$P_t^{PV} = P_t^{PV,unit} P_{PV} \quad (3\text{-}5)$$

The optimization objective is formulated as:

$$maxSCSB = \frac{2SCR \cdot SSR}{SCR + SSR} \quad (3\text{-}6)$$

where SCR is the self-consumption rate and SSR is the self-sufficiency rate.

The PV capacity that maximizes the SCSB is adopted as the installed community PV capacity. The resulting hourly PV generation profile is subsequently used as the renewable energy input for the battery sizing model described in Section 3.3.4.

3.3.4 BESS Sizing

Battery sizing is performed after the PV generation profile has been determined. The selected PV generation profile and community demand profile define the available PV surplus and residual demand, which are used to determine the optimal battery energy and power capacities. Since PV generation has already been fixed through the PV sizing procedure, the battery is sized to increase the utilization of surplus PV generation while improving community self-consumption.

Before battery sizing, the feasible ranges of the self-consumption rate (SCR) and self-sufficiency rate (SSR) are first determined from the selected PV generation profile and the community electricity demand profile. User-defined SCR and SSR targets are required to lie within this feasible range to ensure physical achievability. The validated target values are subsequently incorporated into the battery sizing optimization as operational constraints.

The battery sizing problem is formulated as a Mixed-Integer Linear Programming (MILP) optimization that simultaneously determines the battery capacity and the hourly battery charging and discharging schedules by minimizing the annualized total system cost. During the sizing process, battery charging is restricted to surplus PV generation, while battery discharge is used only to satisfy community demand. Grid charging is not permitted during battery sizing, ensuring that the battery is sized exclusively to increase renewable energy utilization rather than electricity price arbitrage.

The optimization objective is formulated as:

$$minJ^{BESS} = C^{imp} - R^{exp} + C^{cyc} + C^{BESS,cap} + C^{BESS,op} \quad (3\text{-}7)$$

Where C^{imp} is the annual grid import cost, R^{exp} is the annual export revenue, C^{cyc} is the battery cycling cost, $C^{BESS,cap}$ is the annualized battery capital cost and $C^{BESS,op}$ is the annual battery operating cost.

The battery energy capacity is constrained by:

$$0 \leq E^{BESS} \leq E^{BESS}_{max} \tag{3-8}$$

The battery power capacity is linked to energy capacity through a fixed power-to-energy ratio $r_{P/E}$:

$$P^{BESS} = r_{P/E} E^{BESS} \tag{3-9}$$

The battery operation is governed by the state-of-charge (SOC) dynamics, which evolves according to:

$$E^{SOC}_t = E^{SOC}_{t-1} + \eta^{ch} P^{ch}_t \Delta t - \frac{P^{dis}_t \Delta t}{\eta^{dis}} \tag{3-10}$$

subject to the allowable SOC operating limits:

$$SOC_{min} E^{BESS} \leq E^{SOC}_t \leq SOC_{max} E^{BESS} \tag{3-11}$$

Charging and discharging power are limited by the sized battery power capacity:

$$P^{ch,PV}_t \leq P^{BESS} \tag{3-12}$$

$$P^{dis}_t \leq P^{BESS} \tag{3-13}$$

PV charging is limited by the pre-battery PV surplus:

$$P^{ch,PV}_t \leq \max\left(P^{PV}_t - P^{Load}_t, 0\right) \tag{3-14}$$

Battery discharge is limited by the pre-battery residual demand:

$$P^{dis}_t \leq \max\left(P^{Load}_t - P^{PV}_t, 0\right) \tag{3-15}$$

When an SCR or SSR target is active, the battery dispatch is further constrained to satisfy the required annual self-consumed energy:

$$E^{self,direct} + \sum_{t \in \mathcal{T}} P^{dis}_t \Delta t \geq E^{self}_{req} \tag{3-16}$$

where E^{self}_{req} is determined from the validated user-defined SCR or SSR target.

The optimized battery energy and power capacities, together with the selected PV capacity, are subsequently used as fixed design inputs to the operational dispatch model described in Section 3.3.5.

3.3.5 Operational Dispatch

The operational dispatch model determines the optimal hourly operation of the designed PV-BESS system over the simulation horizon. Based on the determined PV capacity and battery capacity in Section 3.3.3 and Section 3.3.4, the model optimizes battery operation and electricity exchange with the grid while satisfying the community energy balance and battery operating constraints.

The operational dispatch is formulated as a Mixed-Integer Linear Programming (MILP) optimization that simultaneously determines the hourly battery charging, battery discharging, state of charge (SOC), grid import, and grid export. Unlike the battery sizing model, the installed PV and battery capacities remain fixed, and only the hourly operational variables are optimized. The dispatch supports the three physical configurations considered in this study, namely PV-only, PV+BESS, and PV+BESS+BTM. Different operational constraints are activated according to the selected system configuration, while the optimization framework remains unchanged.

The dispatch objective minimizes the total operating cost over the simulation period:

$$minJ^{disp} = C^{imp} - R^{exp} + C^{cyc} \tag{3-17}$$

where C^{imp} is the electricity import cost, R^{exp} is the electricity export revenue and C^{cyc} is the battery cycling cost, which is determined based on the degradation model described in Section 3.3.6. Unlike the battery sizing model, PV and battery capital costs are excluded because the system capacities have already been determined.

For each time step, the community energy balance is expressed as:

$$P_t^{PV} + P_t^{imp} + P_t^{dis} = P_t^{Load} + P_t^{ch} + P_t^{exp} \tag{3-18}$$

The battery SOC evolves according to:

$$SOC_t = SOC_{t-1} + \eta^{ch} P_t^{ch} \Delta t - \frac{P_t^{dis} \Delta t}{\eta^{dis}} \tag{3-19}$$

subject to the allowable operating limits:

$$SOC_{min} \leq SOC_t \leq SOC_{max} \tag{3-20}$$

Charging and discharging power are limited by the installed battery power capacity:

$$0 \leq P_t^{ch} \leq P^{BESS} \tag{3-21}$$

$$0 \leq P_t^{dis} \leq P^{BESS} \tag{3-22}$$

The battery is further required to satisfy both the prescribed initial SOC and a cyclic terminal SOC condition to ensure consistent operation over the optimization horizon.

Configuration-specific operational constraints are applied according to the selected physical system. For the PV+BESS configuration, battery charging is restricted to available PV surplus：

$$P_t^{ch,PV} \leq \max\left(P_t^{PV} - P_t^{Load}, 0\right) \tag{3-23}$$

For the PV+BESS+BTM configuration, battery charging from the electricity grid is additionally permitted. To ensure physically consistent energy flows, battery charging from the grid is constrained by:

$$P_t^{ch,G} \leq P_t^{imp} \tag{3-24}$$

Binary decision variables are introduced to prevent simultaneous battery charging and discharging:

$$u_t^{ch} + u_t^{dis} \leq 1 \tag{3-25}$$

and to prohibit simultaneous grid import and export:

$$u_t^{imp} + u_t^{exp} \leq 1 \tag{3-26}$$

These operational constraints ensure physically feasible battery operation while preventing artificial tariff arbitrage between electricity imports and exports. The optimized hourly dispatch schedules are subsequently used for battery degradation analysis and lifecycle economic assessment, as described in Section 3.3.6 and Section 3.3.7.

3.3.6 Battery Degradation

The battery degradation model evaluates the long-term capacity fade of the battery system under the optimized operational dispatch obtained in Section 3.3.5. By quantifying battery degradation over the project lifetime, the model estimates battery lifetime, determines battery replacement events, and provides the battery cycling cost coefficient required for degradation-aware dispatch optimization and lifecycle economic assessment.

Battery degradation is evaluated using the hourly battery discharge profile generated by the operational dispatch model. The annual discharged battery energy is first calculated and converted into equivalent full cycles (EFC) based on the usable battery capacity. Cycle degradation is then estimated from the assumed cycle life and end-of-life capacity fraction, while calendar degradation is applied annually as a proportional capacity loss [55]. The remaining battery capacity is updated year by year throughout the project lifetime. Battery replacement is triggered when the remaining capacity falls below the specified replacement threshold.

The annual discharged battery energy is calculated as:

$$E_{annual}^{dis} = \sum_{t \in \mathcal{T}} P_t^{dis} \Delta t \tag{3-27}$$

where P_t^{dis} is the battery discharge power and Δt is the simulation time step.

The annual equivalent full cycles are calculated as:

$$EFC = \frac{E_{annual}^{dis}}{E^{BESS}(SOC_{max} - SOC_{min})} \tag{3-28}$$

where E^{BESS} is the installed battery energy capacity and $SOC_{max} - SOC_{min}$ is the usable SOC window.

The cycle degradation per equivalent full cycle is determined by:

$$\delta^{cyc} = \frac{1 - \lambda^{EOL}}{N^{cycle}} \tag{3-29}$$

where λ^{EOL} is the end-of-life capacity fraction and N^{cycle} is the cycle life at full depth of discharge.

The annual battery capacity fraction evolves as:

$$\lambda_n = \max\left(\lambda_{n-1} - r^{cal}\lambda_{n-1} - \delta^{cyc} EFC, 0\right) \tag{3-30}$$

where λ_n is the battery capacity fraction at the end of year n, r^{cal} is the annual calendar ageing rate, and $\delta^{cyc} EFC$ is the annual cycle-induced capacity loss.

The battery replacement is triggered when:

$$\lambda_n \leq \lambda^{rep} \tag{3-31}$$

where λ^{rep} is th replacement threhold capacity fraction.

To account for battery degradation during operation, the marginal battery cycling cost is estimated as:

$$c^{cyc} = \frac{C_{kWh}^{BESS} f^{rep}}{N^{cycle}(SOC_{max} - SOC_{min})\sqrt{\eta^{ch}\eta^{dis}}} \quad (3\text{-}32)$$

where C_{kWh}^{BESS} is the BESS capital cost per unit energy capacity, f^{rep} is the replacement cost fraction, and η^{ch} and η^{dis} are the charge and discharge efficiencies, respectively. The resulting marginal cycling cost is incorporated into the operational dispatch objective function described in Section 3.3.5 to account for degradation-related operating costs.

The estimated battery lifetime, replacement schedule, and lifecycle degradation costs are subsequently incorporated into the lifecycle economic assessment presented in Section 3.3.7.

3.3.7 Lifecycle Economic Assessment

The lifecycle economic assessment evaluates the long-term economic performance of the PV-BESS system based on the operational dispatch and battery degradation results obtained in Section 3.3.5 and Section 3.3.6. The assessment quantifies the economic viability of different system configurations by considering initial investment costs, annual operating cashflows, battery replacement costs, and discounted lifecycle cashflows over the project lifetime.

The assessment begins by calculating the capital investment and annual operating expenses of the PV and battery systems according to the selected system capacities. The hourly dispatch results are then converted into electricity-related cashflows by accounting for electricity import costs, electricity export revenues, battery cycling costs, and self-consumption incentives where applicable. The annual operating cashflow is subsequently determined by comparing the simulated system with a baseline scenario without the PV-BESS system.

To evaluate long-term project performance, a two-anchor lifecycle evaluation strategy is developed [56], [57]. Hourly dispatch simulations are performed only for the first project year and the degraded end-of-life system to reduce the computational burden associated with repeated hourly dispatch simulations over the entire project lifetime. The corresponding annual operating cashflows are then linearly interpolated to estimate the operating cashflows for the intermediate project years. Battery replacement costs determined by the degradation model are incorporated into the corresponding project years to construct the complete lifecycle cashflow series. Finally, discounted cashflow analysis is performed to calculate the principal economic indicators, including net present value (NPV), internal rate of return (IRR), and discounted payback period.

The total initial investment is calculated as:

$$C^{CAPEX} = C^{PV} + C^{BESS} \tag{3-33}$$

where C^{PV} is the photovoltaic system capital cost and C^{BESS} is the battery energy storage system capital cost.

The annual operating expenditure is:

$$C^{OPEX} = C^{PV,OPEX} + C^{BESS,OPEX} \tag{3-34}$$

For each simulation time step, the net electricity cost is calculated as:

$$C_t^{ele} = C_t^{imp} - R_t^{exp} + C_t^{cyc} \tag{3-35}$$

where:

$$C_t^{imp} = P_t^{imp} \pi_t^{imp} \Delta t \tag{3-36}$$

$$R_t^{exp} = P_t^{exp} \pi_t^{exp} \Delta t \tag{3-37}$$

where π_t^{imp} grid import electricity price and π_t^{exp} is grid export remuneration.

If self-consumption incentives are considered, the corresponding hourly incentive is calculated as:

$$R_t^{SC} = E_t^{SC} \pi^{SC} \tag{3-38}$$

where E_t^{SC} is the eligible self-consumed PV energy and π^{SC} is the incentive rate for self-consumption.

The annual operating cashflow is obtained by comparing the simulated system with the baseline electricity cost：

$$CF^{op} = S^{bill} + R^{SC} - C^{OPEX} \tag{3-39}$$

where S^{bill} is the annual electricity bill saving relative to the baseline system, calculated as:

$$S^{bill} = C^{baseline} - C^{scenario} \tag{3-40}$$

The project cashflow at the beginning of the project is:

$$CF_0 = -C^{CAPEX} \tag{3-41}$$

For each subsequent project year:

$$CF_n = CF_n^{op} - CF_n^{rep} \tag{3-42}$$

where CF_n^{op} is the annual operating cashflow and CF_n^{rep} is the battery replacement cost when replacement occurs.

The annual operating cashflows are estimated using linear interpolation between the two anchor dispatch simulations:

$$CF_n^{op} = CF_1^{op} + \frac{n-1}{N-1}(CF_N^{op} - CF_1^{op}) \tag{3-43}$$

where N is the project lifetime.

The net present value is calculated as:

$$NPV = \sum_{n=0}^{N} \frac{CF_n}{(1+r)^n} \tag{3-44}$$

where r is the discount rate.

The internal rate of return (IRR) is the discount rate that satisfies:

$$\sum_{n=0}^{N} \frac{CF_n}{(1+r)^n} = 0 \tag{3-45}$$

The discounted payback period is determined as the first year at which the cumulative discounted cashflow becomes non-negative.

3.4 Governance Layer Design

Building upon the physical layer model presented in Section 3.3, this section describes the design of the proposed governance layer, which introduces an energy token mechanism to coordinate renewable energy utilization and benefit distribution within the community. While the physical layer determines how renewable electricity is generated, stored, and shared, the governance layer specifies how the resulting renewable energy contributions are recognized and transformed into governance incentives that encourage long-term participation and coordinated energy management within the community.

The governance layer operates on standardized outputs generated by the physical-layer simulation, ensuring that all governance decisions remain physically grounded. It establishes a complete governance workflow in which renewable energy contributions are converted into energy tokens, distributed among community members according to predefined governance principles, circulated through community activities, and utilized to support demand-side flexibility. Figure 3.4 presents the overall workflow of the proposed energy token mechanism. The following subsections describe each stage of the mechanism in detail, including the physical-governance interface, token

issuance, token allocation, token circulation, and demand response incentives supported by the community treasury design.

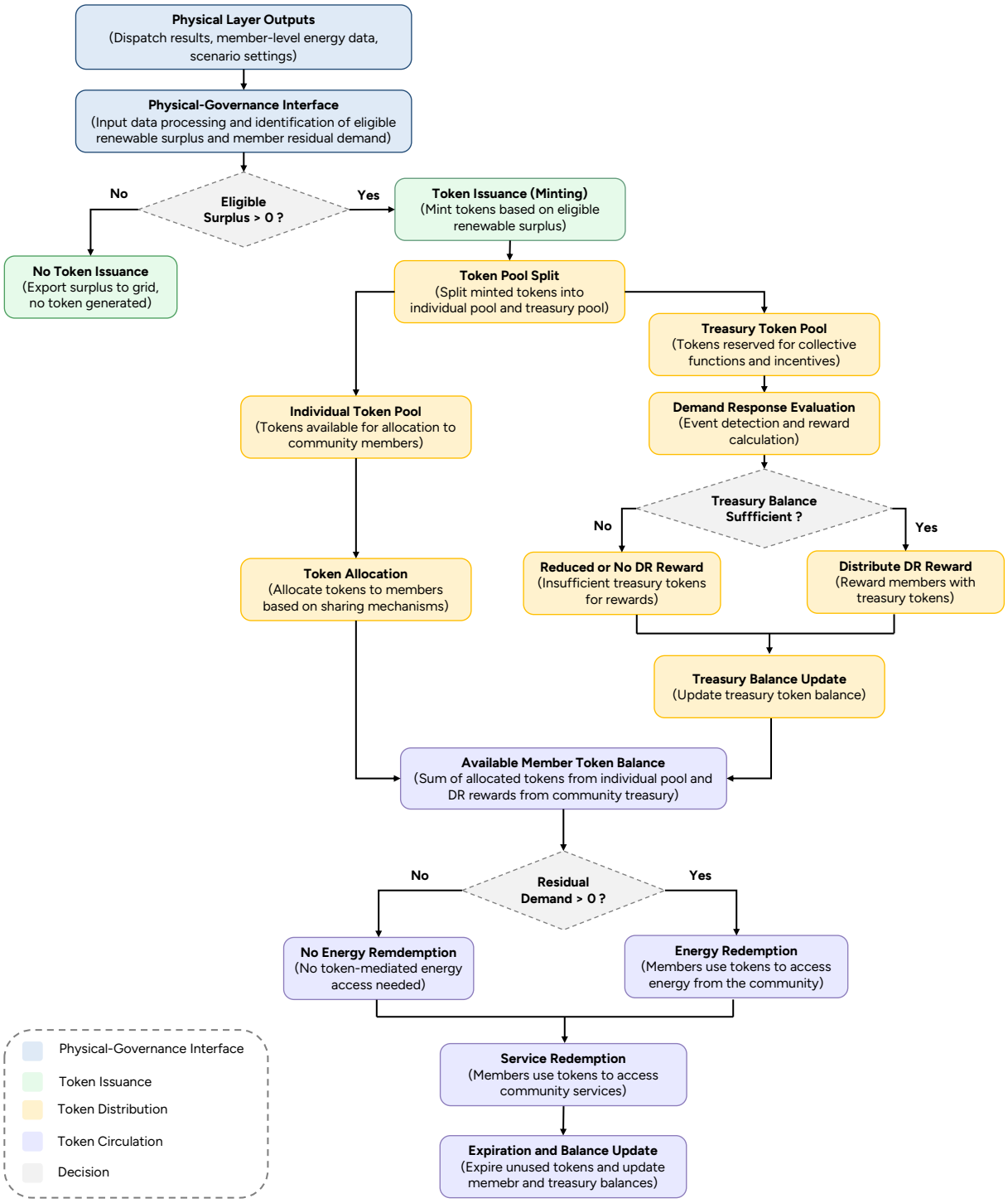


Figure 3.4: Overall workflow of the proposed energy token mechanism

3.4.1 Physical-Governance Interface

The physical–governance interface establishes the connection between the REUnity physical-layer simulation and the proposed energy token mechanism by transforming hourly physical-layer outputs into standardized governance inputs. As illustrated in Figure 3.5, the interface receives community energy flows, member energy data, electricity pricing information, and scenario metadata from the physical-layer simulation. These data are first standardized through temporal synchronization before being reorganized into community-level and member-level governance datasets.

The standardized datasets are subsequently used to derive the governance variables required throughout the token mechanism, including eligible renewable energy surplus, member residual demand, electricity pricing information, battery operating states, and scenario metadata. These governance-ready variables provide the physical basis for the subsequent token issuance, allocation, circulation, and governance evaluation processes.

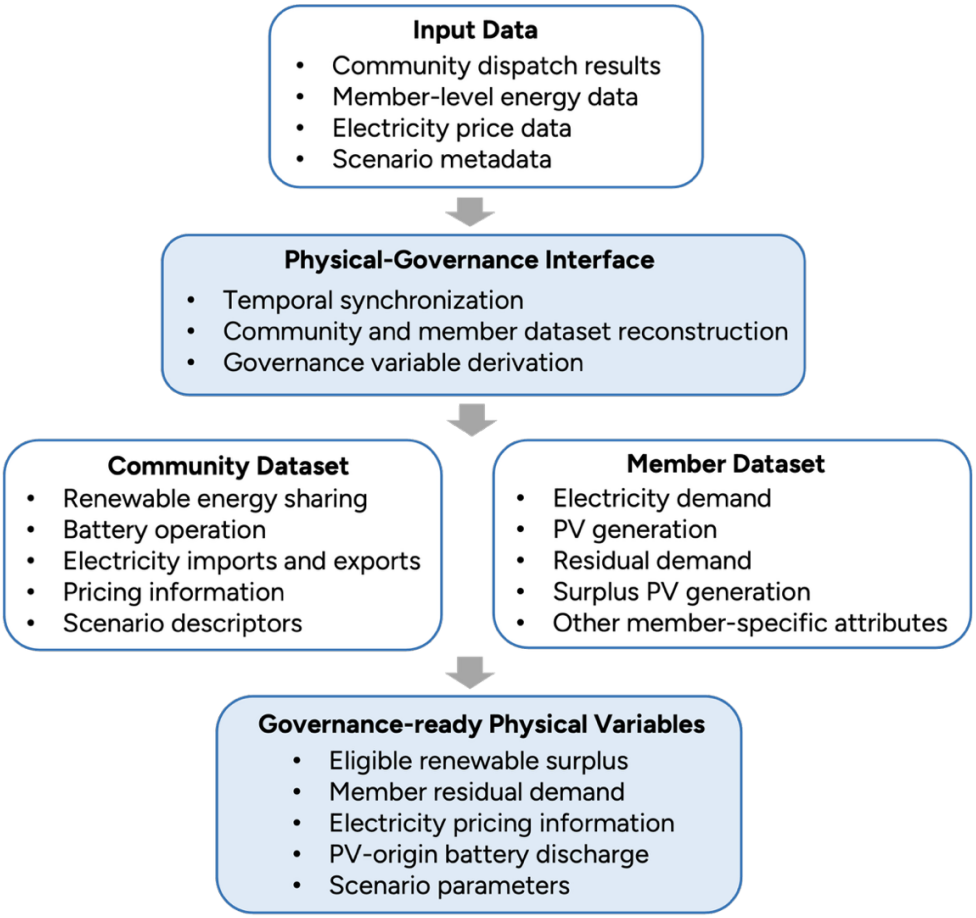


Figure 3.5: Physical-governance data preparation process

3.4.2 Token Issuance

Token issuance is based exclusively on renewable energy that has physically contributed to the community energy supply. Eligible renewable energy consists of two components: immediate renewable energy shared within the community and PV-origin battery discharge supplied to community demand. Battery discharge originating from grid charging is excluded to ensure that token issuance reflects only renewable energy generated within the community. The eligible renewable energy is calculated as:

$$E_t^{eligible} = E_t^{share} + E_t^{PV,BESS} \tag{3-46}$$

where E_t^{share} is the immediate internal renewable energy sharing, and $E_t^{PV,BESS}$ is the photovoltaic-origin battery discharge supplied to community members.

After the eligible renewable energy has been identified, it is converted into newly issued energy tokens using a predefined minting coefficient.

$$T_{mint} = \alpha E_{eligible} \tag{3-47}$$

where T_{mint} is the number of newly minted tokens, α is the token minting coefficient, and $E_{eligible}$ is the eligible renewable energy.

A configurable proportion of the newly minted tokens is reserved for the community treasury before member allocation. The size of the treasury pool is calculated as:

$$T_{treasury} = \tau T_{mint} \tag{3-48}$$

where τ represents the treasury allocation ratio.

3.4.3 Token Allocation

Following token issuance, the newly minted tokens are divided into a community treasury pool and a member token pool. A configurable proportion of the minted tokens is reserved for the treasury, while the remaining tokens are allocated to community members according to the selected allocation mechanism. The number of tokens available for member allocation is determined as:

$$T_{member} = (1 - \tau) T_{mint} \tag{3-49}$$

where T_{member} denotes the member token pool.

The framework supports four configurable member allocation mechanisms [58] to distribute the member token pool among eligible participants.

3.4.3.1 Equal-share allocation

Each member receives an identical share of the member token pool regardless of individual electricity demand or renewable energy production.

$$\omega_i = \frac{1}{N} \tag{3-50}$$

where N denotes the number of community members.

This mechanism emphasizes equality by assigning identical governance rights to all participants.

3.4.3.2 Cascade equal-share allocation

The member token pool is first equally distributed among all members. Members whose allocated tokens exceed their eligible energy demand transfer their unused allocation into a redistribution pool. The remaining pool is then redistributed equally among members with unmet renewable energy demand.

This iterative redistribution continues until either all available tokens have been allocated, or all redeemable renewable energy demand has been satisfied.

Compared with simple equal-share allocation, the cascade mechanism improves renewable energy accessibility while preserving the principle of equal participation.

3.4.3.3 Pro-rate consumption allocation

Tokens are distributed proportionally to each member's eligible electricity demand.

$$\omega_i = \frac{D_i}{\sum_j D_i} \tag{3-51}$$

where D_i denotes the eligible electricity demand of member i.

This mechanism prioritizes members with larger renewable electricity consumption potential.

3.4.3.4 Pro-rate production allocation

Tokens are distributed proportionally to each member's eligible renewable energy contribution.

$$\omega_i = \frac{P_i}{\sum_j P_i} \tag{3-52}$$

where P_i denotes the eligible renewable energy contribution of member i.

This mechanism rewards renewable energy producers according to their contribution to the community energy supply.

Once the allocation weights have been determined, the token allocation for each member is calculated as:

$$T_i = \omega_i T_{member} \tag{3-53}$$

where T_i denotes the number of tokens allocated to member i.

3.4.4 Community Treasury and Demand Response Incentives

The community treasury manages the proportion of newly minted tokens reserved during the token pool split. Rather than being allocated directly to community members, treasury-held tokens are retained as a shared governance resource and are subsequently redistributed through configurable demand-response incentive mechanisms.

The framework supports three configurable demand-response modes: no demand response, peak reduction, and surplus absorption. Under the peak reduction strategy, demand-response events correspond to periods of high community electricity imports, whereas under the surplus absorption strategy, events correspond to periods with high eligible renewable energy availability, creating opportunities to increase local renewable energy utilization. When no demand-response strategy is selected, treasury-funded rewards are not activated.

Rather than applying fixed thresholds, the framework identifies demand-response events using scenario-relative quantile-based event detection, allowing the same governance mechanism to operate consistently across different physical configurations despite variations in electricity demand and renewable energy generation. For each simulation timestep, an event indicator is determined as:

$$S^t = \begin{cases} 1, X^t \geq Q_q(X) \\ 0, otherwise \end{cases} \tag{3-54}$$

where X^t denotes either the community import or eligible renewable surplus depending on the selected demand-response strategy, and $Q_q(X)$ represents the corresponding quantile threshold. In the surplus-absorption mode, the event signal is based on eligible renewable energy rather than grid export, reflecting an opportunity for local renewable energy utilization within the token mechanism.

During identified demand-response events, a predefined share of each member's electricity demand is treated as flexible, and the participating share of this flexible demand is determined by the configured participation ratio. Members contributing

flexible demand become eligible to receive treasury-funded rewards. The total reward distribution is constrained by the available treasury balance. When the available treasury balance is insufficient to satisfy the total reward requirement, reward payments are proportionally scaled while preserving the relative reward shares among eligible members.

Demand response is represented through rule-based temporal load shifting. For each participating member, a configurable share of hourly electricity demand is treated as flexible. Under peak reduction, flexible demand is shifted from an identified event hour to the nearest eligible non-event hour within the same day. Under surplus absorption, flexible demand is shifted from the nearest eligible non-event hour into the identified surplus-event hour. The shifting process conserves total electricity demand rather than curtailing consumption.

In the evaluated scenarios, 10% of household electricity demand is assumed to be flexible, with a participation ratio of 50%. Load shifting is restricted to the period between 06:00 and 23:00 and does not occur across calendar days. The model selects the nearest eligible hour and does not impose an additional fixed limit on the shifting distance within this daily window.

To maintain a consistent timestep-based circulation process, rewards earned during timestep t are credited at the beginning of the subsequent timestep:

$$T_{i,DR}^{t} \rightarrow B_i^{t+1} \tag{3-55}$$

This delayed settlement separates flexibility provision from subsequent token utilization while preserving the temporal consistency of the governance mechanism.

3.4.5 Token Circulation

The dynamic token circulation process models the evolution of member token balances throughout the simulation. At each simulation timestep, token balances are sequentially updated according to renewable energy redemption, community service redemption, token expiration, and balance carry-over. This state-based representation enables the complete lifecycle of energy tokens to be tracked while maintaining consistency with the hourly operation of the renewable energy community.

3.4.5.1 Update the Opening Token Balance

At the beginning of each simulation timestep, the available token balance is determined by combining the remaining balance from the previous timestep, newly allocated

member tokens, and delayed demand-response rewards received from the previous timestep. The opening token balance is calculated as:

$$B_i^t = B_i^{t-1} + T_{i,alloc}^t + T_{i,DR}^{t-1} - T_{i,exp}^t \quad (3\text{-}56)$$

where B_i^t is the opening token balance of member i at timestep t, $T_{i,alloc}^t$ denotes newly allocated tokens, $T_{i,DR}^{t-1}$ represents demand response rewards earned in the previous timestep and received at the current timestep, and $T_{i,exp}^t$ denotes tokens that expire at timestep t.

3.4.5.2 Redeem Tokens for Renewable Energy

Members may redeem tokens to obtain access to eligible renewable electricity. The amount of renewable electricity that can be accessed is jointly constrained by the member's residual electricity demand, the available token balance, and the remaining eligible renewable electricity within the community. The redeemable renewable electricity is therefore calculated as:

$$E_{i,access}^t = \min\left(D_{i,res}^t, \frac{B_i^t}{\gamma}, E_{eligible,remain}^t\right) \quad (3\text{-}57)$$

where $D_{i,res}^t$ is the member's residual electricity demand, $E_{eligible,remain}^t$ represents the remaining eligible renewable energy available within the community, and γ denotes the token-to-energy exchange coefficient.

The corresponding tokens are permanently redeemed and removed from the member balance:

$$T_{i,energy}^t = \gamma E_{i,access}^t \quad (3\text{-}58)$$

3.4.5.3 Redeem Tokens for Community Services

Following renewable energy redemption, members may voluntarily exchange part of their remaining token balance for community services. The framework supports configurable service redemption frequencies, including hourly, daily, weekly, and monthly intervals. At each service redemption event, members may redeem a predefined proportion of their remaining token balance, subject to the available balance and the maximum redeemable service capacity. The redeemed service tokens are calculated as:

$$T_{i,service}^t = \min\left(\rho_s B_i^{t,after\ energy}, C_i^t\right) \quad (3\text{-}59)$$

where ρ_s is the service spending ratio and C_i^t represents the maximum redeemable service capacity.

Redeemed service tokens are permanently removed from circulation and subsequently converted into community service value during the economic valuation process described in Section 3.5.

3.4.5.4 Update the Closing Token Balance

After all redemption activities have been completed, the remaining token balance is transferred to the subsequent simulation timestep. Tokens exceeding their configured lifetime are permanently removed from circulation according to a first-in-first-out (FIFO) expiration policy. The closing token balance is calculated as:

$$B_i^{t,close} = B_i^t - T_{i,energy}^t - T_{i,service}^t \tag{3-60}$$

where $T_{i,energy}^t$ denotes the tokens redeemed for renewable electricity access and $T_{i,service}^t$ denotes the tokens redeemed for community services.

The closing balance becomes the opening balance of the following timestep:

$$B_i^{t,close} \rightarrow B_i^{t+1} \tag{3-61}$$

thereby forming a continuous state-transition process throughout the simulation horizon.

3.5 Scenario Design and Evaluation Framework

The proposed governance framework is evaluated under multiple physical configurations and governance parameter combinations to investigate the interaction between community energy performance and governance design. The evaluation framework consists of three components: physical configurations representing different renewable energy system designs, governance design variables defining alternative token mechanisms, and a performance assessment framework used to compare the resulting governance outcomes.

3.5.1 Physical Configurations

Three representative physical configurations are considered, representing successive levels of renewable energy integration within the community and providing the physical operating conditions for the governance evaluation.

The evaluated configurations are:

1) PV-only, in which photovoltaic generation is integrated without battery storage.
2) PV+BESS, where a battery energy storage system is integrated to increase local renewable energy utilization through energy storage and time-shifted dispatch.
3) PV+BESS+BTM, where the battery system additionally supports behind-the-meter grid charging for price-based energy arbitrage.

All governance evaluations are performed under an electricity price scenario combining hourly retail electricity pricing with spot-market feed-in remuneration. By maintaining a consistent electricity pricing environment across all experiments, the effects of different governance configurations can be evaluated independently of variations in market conditions.

3.5.2 Governance Design Space

To investigate the influence of governance design on community performance, a systematic governance design space is constructed by varying the principal configurable parameters of the proposed token mechanism while maintaining the same physical operating conditions. These parameters represent the key governance decisions affecting token distribution, treasury operation, token circulation and value realization. The evaluated governance parameters and their corresponding design space are summarized in Table 3.1.

Table 3.1: Governance design space

Governance parameter	**Design space**
Allocation mechanism	Equal share, Cascade equal share, Pro-rate consumption, Pro-rate production
Treasury allocation ratio	0%, 10%, 20%, 30%, 40%, 50%
Token lifetime	1 day, 1 week, 1 month, 3 months, 1 year
Demand response mode	No DR, Peak reduction, Surplus absorption

Service spending ratio	0%, 5%, 10%, 20%, 30%, 40%
Service spending frequency	Hourly, Daily, Weekly, Monthly

Each governance parameter is first evaluated individually while the remaining parameters are held constant to isolate its influence on governance performance. Based on these parameter-specific analyses, refined parameter settings are subsequently combined to construct the integrated governance configurations used for the final governance assessment. This sequential exploration enables both the individual effects of each governance assessment and their combined performance to by systematically evaluated.

3.5.3 Performance Assessment

A multidimensional performance assessment framework is applied to examine governance performance from complementary perspectives. Rather than relying on a single optimization objective, the framework evaluates renewable energy utilization, economic value creation, distributional fairness, token circulation, and integrated governance performance. Together, these indicators, summarized in Table 3.2, provide a comprehensive basis for comparing alternative governance configurations and identifying balanced governance designs.

Table 3.2: Governance performance indicators

Category	**Performance indicators**
Renewable energy utilization	Energy Access Rate
Economic performance	Realized Energy Value, Realized Service Value, Total Realized Economic Value
Distributional fairness	Jain's Fairness Index, Gini Coefficient, Coefficient of Variation, Minimum-to-Maximum Ratio
Token circulation	Token Utilization Rate, Token Expiration Rate, Treasury Utilization Rate
Integrated governance performance	Activation Score, Overall Score

3.5.3.1 Renewable Energy Utilization

Renewable energy utilization is evaluated using the Energy Access Rate, defined as:

$$R_{enery\ access} = \frac{E_{access}}{E_{eligible}} \tag{3-62}$$

where E_{access} denotes renewable energy successfully accessed through token redemption, and $E_{eligible}$ represents total renewable energy eligible for token issuance. A higher energy access rate indicates that a larger proportion of eligible renewable energy is ultimately utilized through token redemption within the proposed governance framework.

3.5.3.2 Economic Performance

Economic performance is evaluated through the realized economic benefits generated by token utilization. Three complementary indicators are considered: Realized Energy Value, Realized Service Value, and Total Realized Economic Value.

Unlike conventional digital currencies, the proposed energy tokens do not possess an intrinsic monetary value. Instead, economic value is realized only when tokens are redeemed for renewable electricity or community services. Renewable energy redemption is valued according to the avoided electricity expenditure achieved through community energy access, while community service redemption is evaluated using a predefined service reference value. The Total Realized Economic Value is therefore defined as:

$$V_{realized} = V_{energy} + V_{service} \tag{3-63}$$

where V_{energy} denotes the realized economic value generated through renewable electricity redemption, and $V_{service}$ represents the realized economic value obtained through community service redemption. Together, these indictors quantify the overall economic outcomes generated by the proposed governance mechanism over the simulation period.

3.5.3.3 Distributional Fairness

Distribution fairness is evaluated using four complementary statistical indicators [58], taking token-mediated renewable-energy access as the realized benefits.

The **Jain's Fairness Index** evaluates the overall equality of benefit distribution,

$$J = \frac{(\sum_{i=1}^{N} x_i)^2}{N \sum_{i=1}^{N} {x_i}^2} \tag{3-64}$$

where x_i denotes the realized benefit received by member i, and N is the total number of community members.

The Jain's Fairness Index ranges from 0 to 1, with higher values indicating a more equitable distribution.

The **Gini Coefficient** measures the degree of inequality within the benefit distribution,

$$G = 1 - \frac{\sum_{i=1}^{N}(2i - N - 1)x_i}{N \sum_{i=1}^{N} x_i} \tag{3-65}$$

where x_i denotes ordered realized benefit of member i.

The Gini Coefficient ranges from 0 to 1, where lower values indicate greater equality.

The **Coefficient of Variation (CV)** evaluates the relative dispersion of realized benefits,

$$CV = \frac{\sigma}{\bar{x}} \tag{3-66}$$

where σ is the standard deviation of realized benefits and $\bar{x}$ is their mean.

Smaller CV values indicate more homogeneous benefit distribution among members.

The **Minimum-to-Maximum Ratio** measures the disparity between the least and most benefited members,

$$MinMax = \frac{\min(x_i)}{\max(x_i)} \tag{3-67}$$

MinMax values closer to one indicate that governance benefits are distributed more uniformly across the community.

Together, these indicators provide complementary perspectives on governance equity. Among them, Jain's Fairness Index is adopted as the primary fairness indicator because of its intuitive interpretation, bounded range, and suitability for comparing alternative governance configurations. For the integrated governance assessment, Jain's Fairness Index is therefore reported as the Fairness Score.

3.5.3.4 Token Circulation

Token circulation is evaluated using the Token Utilization Rate, Token Expiration Rate, and Treasury Utilization Rate.

The **Token Utilization Rate** is defined as:

$$R_{token\ utilization} = \frac{T_{energy} + T_{service}}{T_{minted}} \tag{3-68}$$

where T_{energy} denotes the tokens redeemed for renewable electricity access, $T_{service}$ denotes the tokens redeemed for community services, and T_{minted} is the total number of issued tokens.

The **Token Expiry Rate** represents the proportion of issued tokens that expire before being redeemed, which is calculated as:

$$R_{token\ expiry} = \frac{T_{expired}}{T_{minted}} \tag{3-69}$$

where $T_{expired}$ denotes the expired tokens.

Treasury operation is evaluated using the **Treasury Utilization Rate**:

$$R_{treasury\ utilization} = \frac{T_{DR}}{T_{treasury}} \tag{3-70}$$

where T_{DR} denotes the treasury tokens distributed through demand-response incentives and $T_{treasury}$ represents the total treasury token allocation. A higher Treasury Utilization Rate indicates that a larger proportion of treasury resources is actively deployed to support community demand-response activities.

3.5.3.5 Integrated Governance Performance

Because governance performance involves multiple potentially competing objectives, two composite indictors are introduced to facilitate governance comparison.

The **Activation Score** jointly evaluates renewable energy utilization and token circulation,

$$A = \frac{R_{energy\ access} + R_{token\ utilization}}{2} \tag{3-71}$$

where higher values indicate that the governance mechanism both improves renewable energy utilization and promotes active token circulation.

To support governance design exploration and parameter sensitivity analysis, the **Overall Score** is defined as,

$$O = 0.4A + 0.4J - 0.2R_{token\ expiry} \tag{3-72}$$

where A denotes the Activation Score, J denotes Jain's Fairness Index, and $R_{token\ expiry}$ denotes the Token Expiry Rate.

The weighting coefficients emphasize renewable energy utilization and equitable benefit distribution while penalizing excessive token expiration. The Overall Score is adopted as the primary comparative metric for ranking alternative governance configurations and identifying balanced governance designs.

4 Results

4.1 Physical Layer Design

An appropriate physical configuration of the energy community was first established using the REUnity modelling framework. The physical layer was designed through a sequential process consisting of PV sizing followed by BESS sizing. The battery was sized based on the renewable generation profile obtained from the selected PV capacity, ensuring that the storage system sizing reflected the renewable generation and demand characteristics of the selected community.

The simulated monthly PV yield profile used for subsequent system design is shown in Figure 4.1. As expected for the Nordic climate, PV generation exhibited a pronounced seasonal pattern, with electricity generation concentrated between spring and early autumn and substantially lower yields during the winter months. This PV yield profile provided the renewable resource basis for the subsequent PV sizing process and defined the seasonable availability of renewable electricity throughout the subsequent system analysis.

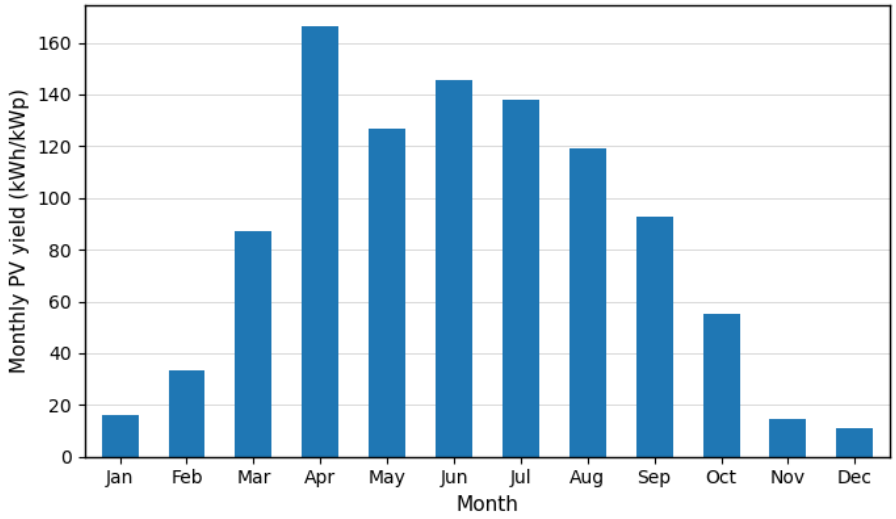


Figure 4.1: Monthly PV yield profile

The PV capacity was optimized by maximizing the self-consumption and self-sufficiency balance (SCSB) objective with the maximum installable capacity constrained by the available rooftop area of the community. As shown in Figure 4.2, the SCSB objective increased with increasing PV capacity until the maximum rooftop utilization potential was reached. Consequently, the rooftop capacity limit of 992.225 kWp was adopted as the final PV installation for the subsequent analysis.

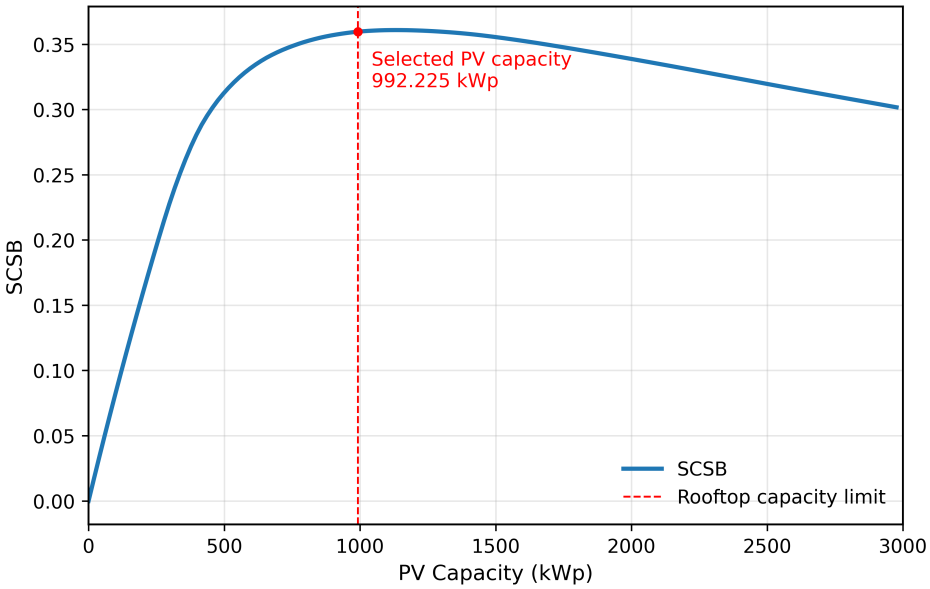


Figure 4.2: Variation of the SCSB objective with PV capacity

Based on the selected PV capacity, the hourly PV generation profile was combined with the community electricity demand profile to determine the feasible range of SCR and SSR. For SCR, the feasible range is from 59.83% to 100%. For SSR, the feasible range is between 25.71% and 42.97%. Target values of SCR = 70% and SSR = 30% were adopted as representative operating targets. These targets are consistent with the physical characteristics of the selected community and lie well within the achievable operating range defined by the available PV generation and community demand profile.

Subsequently, the BESS capacity was determined using the MILP-based battery sizing model. The power capacity of 335.215 kW and the energy capacity of 670.430 kWh were recommended as the optimal solution. The resulting PV-BESS configuration was adopted as the physical foundation for the performance evaluation presented in Section 4.2 and governance analyses in the subsequent sections.

4.2 Physical Layer Performance

Following the physical-layer design presented in Section 4.1, the selected physical configurations were further evaluated using the MILP-based operational optimization model to determine the optimal hourly dispatch of photovoltaic generation, battery storage and grid interaction. The resulting hourly energy flows were then used to assess the energy and economic performance of each physical configuration.

4.2.1 Energy Performance

The energy performance was evaluated in terms of annual grid exchange and renewable energy utilization. Figure 4.3 compares the annual grid import and export under the three physical configurations, while Figure 4.4 presents the corresponding self-consumption ratio (SCR) and self-sufficiency ratio (SSR).

Compared with the PV-only configuration, integrating a community battery substantially reduced both grid electricity import and export. Annual grid import was reduced from 1.73 GWh for the PV-only configuration to 1.68 GWh with the PV+BESS configuration, while annual grid export decreased from 401.9 MWh to 347.0 MWh. Compared with the PV+BESS configuration, the PV+BESS+BTM configuration resulted in a slight increase in both grid electricity import and export. This is because the BTM operational strategy allows the battery to be charged from the grid during low-price periods in addition to storing surplus PV generation, thereby increasing electricity exchanged with the grid while providing greater operational flexibility for price-based energy management. These results indicate that the battery effectively shifted excess PV generation to periods of higher electricity demand, thereby increasing local renewable energy utilization and reducing grid dependence. Allowing grid charging slightly increases grid electricity exchange because of additional price-driven charging operations.

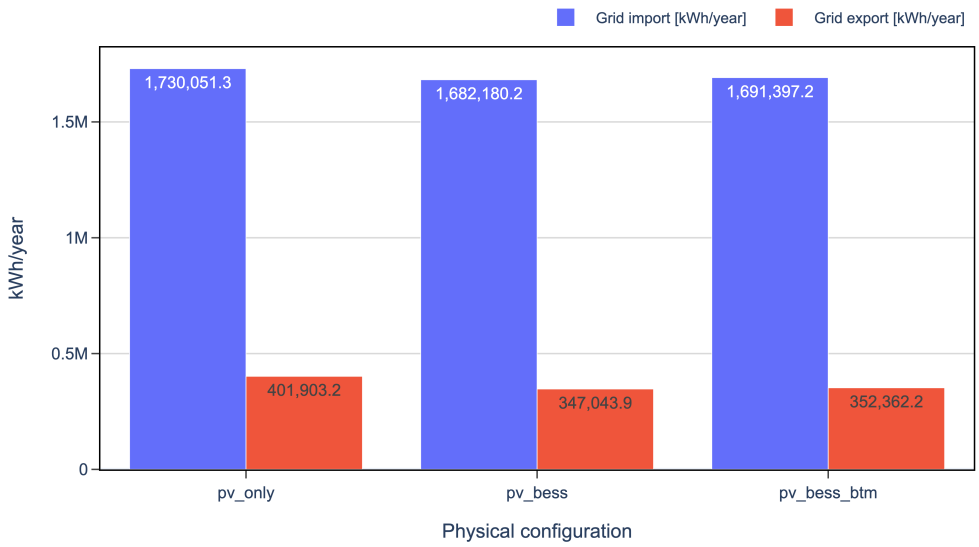


Figure 4.3: Annual grid import and export under three physical configurations

Concerning renewable energy utilization indicators, the addition of battery storage also improved the system performance. The SCR increased from 59.8% for the PV-only configuration to 64.9% for PV+BESS, while only a marginal further increase to 65.3% was observed for PV+BESS+BTM. Similarly, the SSR increased from 25.7% to 27.9%

and 28.0%, respectively. These results indicate that battery storage effectively enhances the utilization of locally generated photovoltaic electricity by shifting surplus PV generation to periods of electricity demand. In contrast, allowing grid charging provides only marginal additional improvements in terms of renewable energy utilization, which indicates that its primary benefit lies in operational flexibility rather than increasing the use of renewable electricity.

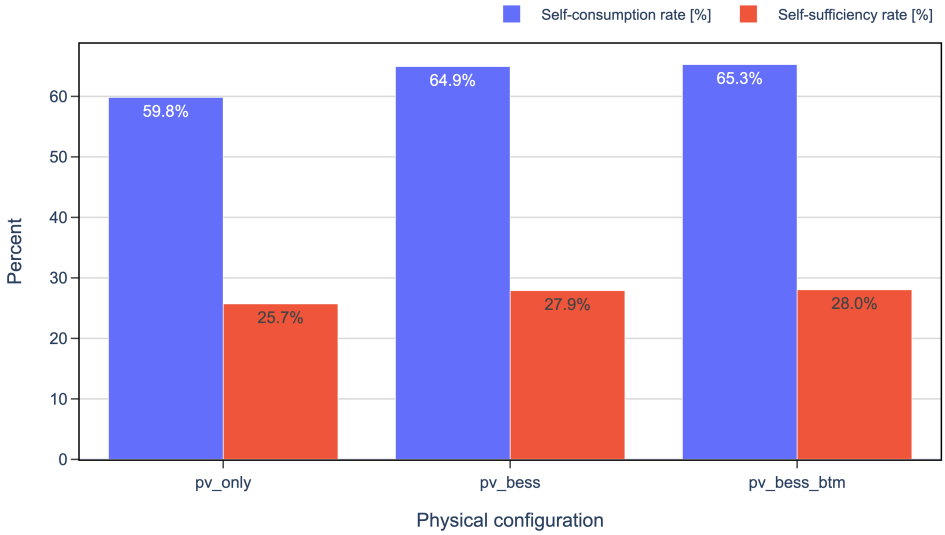


Figure 4.4: SCR and SSR under three physical configurations

Overall, the results demonstrate that the integration of battery storage enhances the operational performance of the energy community by reducing grid dependence and increasing the utilization of locally generated renewable energy. The corresponding economic consequences of these operational characteristics are analyzed in the following section.

4.2.2 Economic Performance

The economic performance of the three physical configurations was evaluated based on their annual electricity bills and lifecycle economic indicators. Figure 4.5 compares the annual electricity bills and corresponding annual bill savings under each physical configuration, while Table 4.1 summarizes the key lifecycle economic performance indicators.

As shown in Figure 4.5, the integration of battery storage reduced the annual electricity bill and increased the annual bill savings compared with the PV-only configuration. The annual electricity bill decreased from 222,424 EUR for the PV-only configuration to 218,778 EUR for the PV+BESS configuration and further to 217,056 EUR for the PV+BESS+BTM configuration. Consequently, the corresponding annual bill savings

increased from 118,480 EUR to 122,126 EUR and 123,848 EUR, respectively. The additional improvement achieved by the PV+BESS+BTM configuration was relatively small, which indicates that allowing grid charging provides only limited additional operational cost savings under the selected pricing scenario.

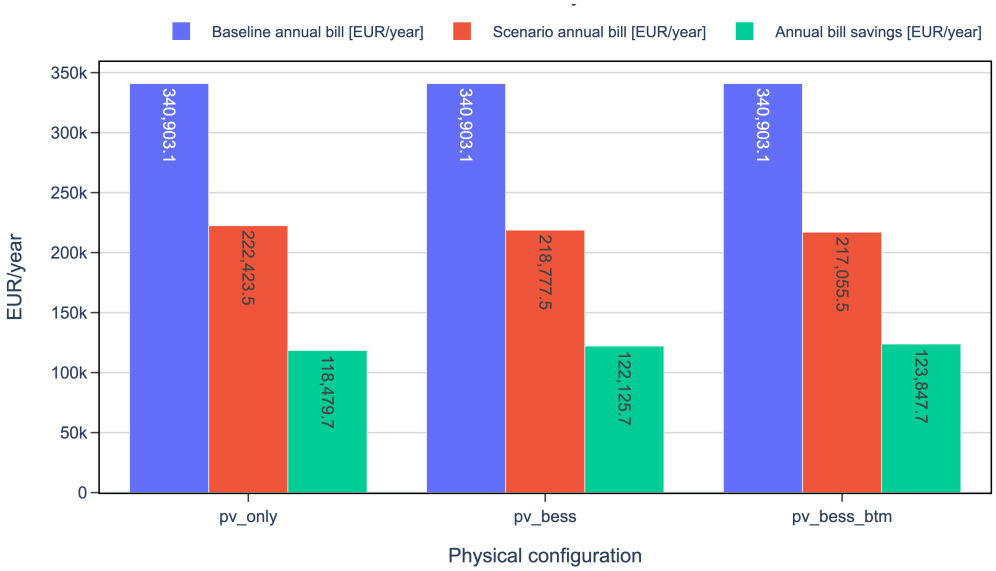


Figure 4.5: Annual electricity bills under three physical configurations

However, the lifecycle economic assessment presented in Table 4.1 reveals a different trend. While both battery-assisted configurations achieved higher annual operating savings, the inclusion of battery storage substantially increased the initial capital investment, from 595,335 EUR for the PV-only configuration to 897,028 EUR for both battery-assisted configurations. As a result, the PV-only configuration achieved the highest lifecycle economic performance, with an NPV of 839,077 EUR, an IRR of 15.89%, and a discounted payback period of 7.50 years. In comparison, the PV+BESS and PV+BESS+BTM configurations yielded considerably lower NPVs of 215,889 EUR and 222,755 EUR, respectively, with corresponding IRRs of approximately 7.2% and discounted payback periods close to 19 years.

Battery degradation also influenced the long-term economic performance of the battery-assisted configurations. The estimated battery lifetime was approximately 9 years for the PV+BESS configuration and 8 years for the PV+BESS+BTM configuration, resulting in three battery replacements over the 30-year project lifetime for both cases. The shorter estimated battery lifetime observed under the PV+BESS+BTM configuration reflects the additional cycling associated with grid charging and discharging, although its overall economic performance remained slightly superior to that of the PV+BESS configuration due to the increased operational savings.

Table 4.1: Lifecycle economic performance under three physical configurations

Indicator	PV only	PV+BESS	PV+BESS+BTM
Initial CAPEX (EUR)	595,335	897,028	897,028
Annual bill savings (EUR/year)	118,480	122,126	123,848
NPV (EUR)	839,077	215,889	222,755
IRR (%)	15.89	7.24	7.26
Discounted payback (years)	7.50	19.13	18.87
Battery lifetime (years)	—	9	8
Number of battery replacements	—	3	3

Overall, these results demonstrate a clear trade-off between operational and lifecycle economic performance. Battery storage improves annual operational cost savings, whereas the associated capital investment and battery replacement costs substantially reduce the overall lifecycle economic benefits under the assumed economic conditions.

4.3 Governance Layer Design

The governance layer was further designed based on the hourly energy flows generated by the REUnity modelling framework. The governance design space was systematically explored across the three physical configurations by varying the key governance parameters, including the allocation mechanism, treasury allocation ratio, token lifetime, demand-response mode, service spending ratio, and service spending frequency. Based on the overall governance performance, the recommended governance configuration was subsequently identified for the further evaluation.

4.3.1 Design Space Exploration

The governance design process consisted of a series of design space explorations, in which each parameter was evaluated individually while the remaining parameters were fixed. The following sections present the influence of each parameter on the overall governance performance before final governance configuration is identified through the integrated assessment.

4.3.1.1 Allocation mechanism

The allocation mechanism determines how the individually distributable token pool is assigned among community members and therefore directly affects both token activation and distributional fairness. Four mechanisms were explored, namely equal share, cascade equal share, pro-rate consumption and pro-rate production. These mechanisms represent different principles for distributing governance incentives among community members.

Figure 4.6 compares the overall governance performance achieved by the four allocation mechanisms. Cascade equal share consistently produced the highest overall score across all three physical configurations, ranging from 0.642 to 0.647. Its advantage over equal share and pro-rate consumption was relatively small, with the overall scores differing by less than 0.03, but the ranking remained stable across the three physical configurations. In contrast, pro-rate production resulted in substantially lower overall performance, particularly under the PV-only configuration where the overall score decreased to 0.274. This is mainly because pro-rate production concentrates tokens on members with larger renewable-energy contributions, while these members may have relatively limited residual demand and therefore fewer opportunities to redeem tokens through renewable-energy access. Under the PV-only configuration, the temporal mismatch between surplus generation and residual demand cannot be buffered by storage, which further constrains token activation and reduces the overall governance score. When BESS is introduced, storage partly relaxes this temporal mismatch by shifting renewable energy to periods with residual demand, which improves token redemption opportunities and substantially increases the performance of the pro-rate production mechanism.

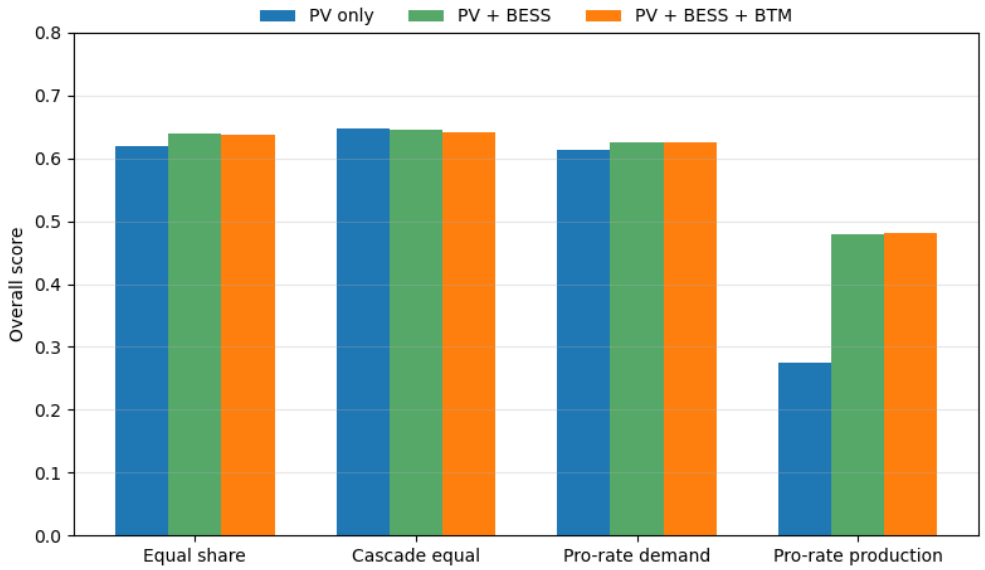


Figure 4.6: Overall governance performance across allocation mechanisms

The activation-fairness trade-off presented in Figure 4.7 provides further insight into these differences. Pro-rate consumption achieved the highest activation scores (0.900-0.949), indicating stronger token circulation and renewable energy access, but this was accompanied by a noticeable reduction in distributional fairness (0.603-0.679). In contrast, equal share maintained the highest balance between the two objectives, achieving relatively high activation (0.812–0.840) while preserving strong fairness (0.767–0.829). Cascade equal share further increased activation (0.880-0.947) relative to equal share with only a modest reduction in fairness (0.695-0.739), explaining its superior overall performance. By comparison, pro-rate production consistently exhibited the lowest activation performance (0.386-0.497) despite maintaining relatively high fairness (0.857-0.862) under the battery-assisted configurations, indicating that prioritizing renewable contributors substantially limits community-wide energy access within the proposed governance framework.

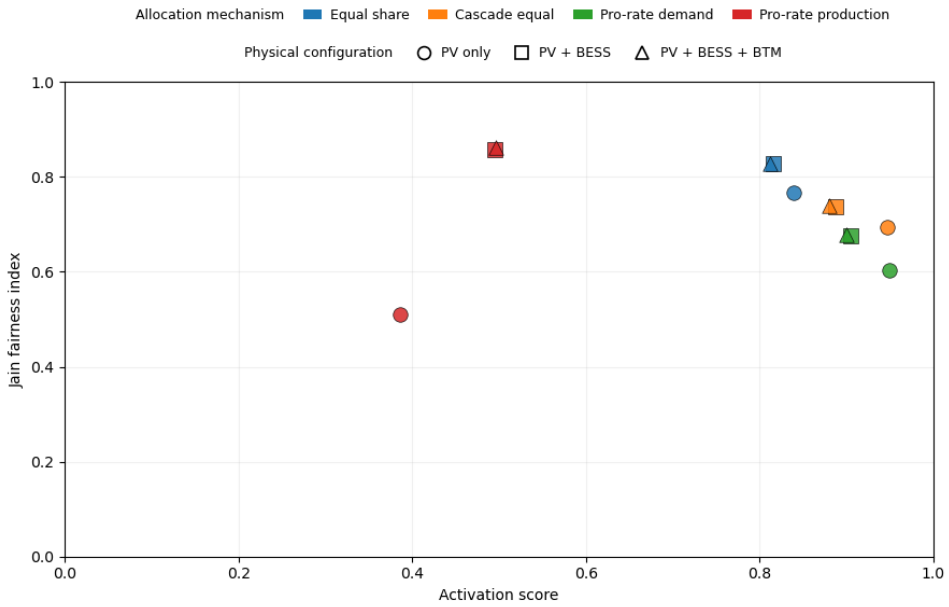


Figure 4.7: Activation-fairness trade-off across four allocation mechanisms

In general, the allocation mechanism primarily influences the balance between token activation and distributional fairness, highlighting the importance of considering multiple governance objectives when selecting an appropriate allocation strategy.

4.3.1.2 Treasury Allocation Ratio

The treasury allocation ratio determines the proportion of newly minted tokens reserved for the community treasury, with the remaining tokens distributed directly to individual members. While treasury tokens are used to finance demand-response incentives in the proposed governance framework, increasing the treasury allocation

simultaneously reduces the size of the individual token pool and may therefore influence both token circulation and governance performance.

Figure 4.8 compares the overall governance performance under different treasury allocation ratios. Across all three physical configurations, the overall score decreased progressively as the treasury allocation ratio increased. For the PV-only configuration, the overall score decreased from 0.652 at a treasury allocation ratio of 0 to 0.562 at a ratio of 0.5. Similarly, the overall scores of the PV+BESS and PV+BESS+BTM configurations declined from 0.676 to 0.581 and from 0.674 to 0.580, respectively. The battery-assisted configurations consistently achieved slightly higher overall scores than the PV-only configuration, but the influence of the treasury allocation ratio remained similar across all three physical configurations. Although a treasury allocation ratio of 0 achieved the highest overall score across the three physical configurations, it does not establish a community treasury and therefore cannot support treasury-funded demand-response incentives. Consequently, it was not considered a practical governance configuration for subsequent analysis.

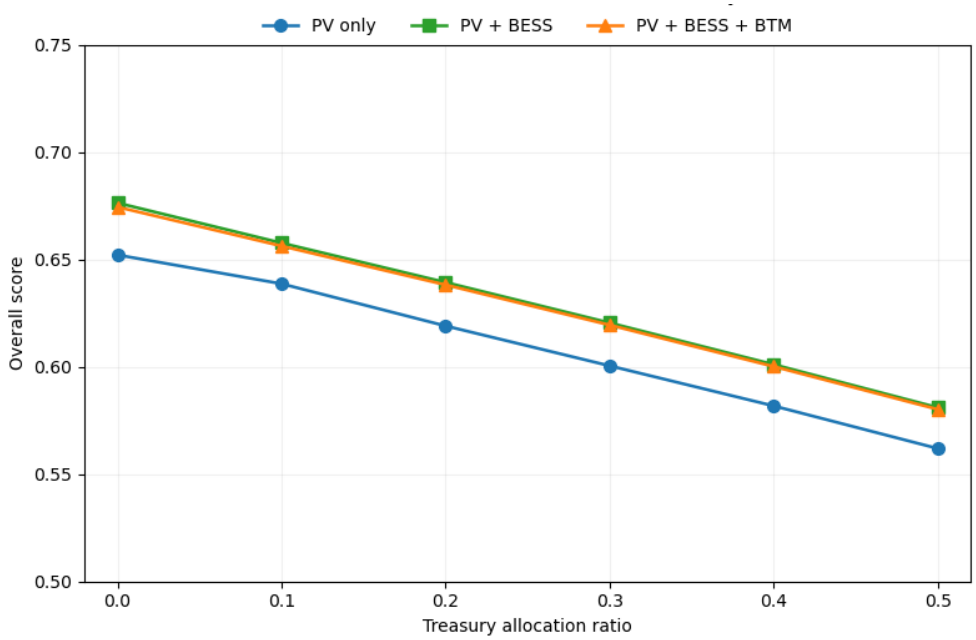


Figure 4.8: Overall performance under different treasury allocation ratios

Figure 4.9 further explains this trend by examining the utilization of treasury-funded demand-response rewards. At a treasury allocation ratio of 0.1, nearly all treasury tokens were redistributed through demand-response incentives under the PV-only configuration, resulting in a treasury utilization rate of 99.4%. However, as the treasury allocation ratio increased, treasury utilization declined rapidly, falling to 28.5% at a ratio of 0.5. Similar reductions were observed for the PV+BESS configuration (from 89.9% to 18.2%) and the PV+BESS+BTM configuration (from 87.3% to 17.7%). These results

indicate that treasury income increased substantially faster than the demand-response rewards distributed under the current governance mechanism. Consequently, allocating a larger proportion of newly minted tokens to the treasury reduced the individual token pool without providing a corresponding increase in treasury-funded incentives, thereby lowering the overall governance performance.

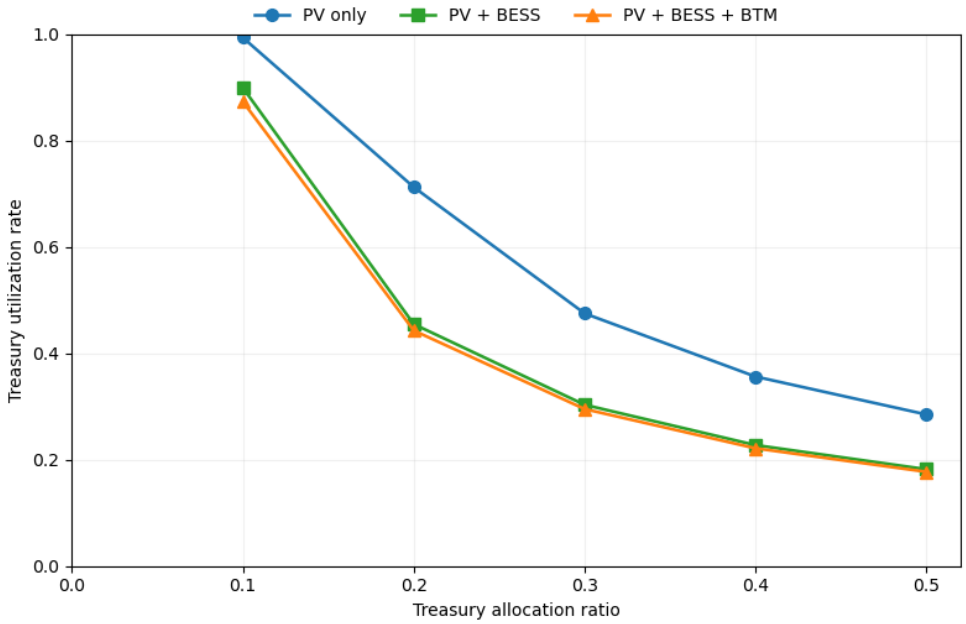


Figure 4.9: Utilization of DR rewards under different allocation ratios

Overall, these results suggest that a modest treasury reserve is sufficient to support the demand-response mechanism under the current modelling assumptions. Further increasing the treasury allocation reduces the individual token pool without proportionally increasing treasury-funded reward expenditure, thereby weakening overall governance performance. This pattern partly reflects the prescribed DR response in the current model, where additional treasury resources do not induce greater participation or flexibility provision.

4.3.1.3 Token Lifetime

The token lifetime determines the period during which minted tokens remain valid before they expire. A longer lifetime provides members with more opportunities to redeem tokens for renewable energy access or community services, whereas a shorter lifetime increases the likelihood of token expiration and may reduce the overall circulation efficiency.

Figure 4.10 illustrates the influence of token lifetime on token utilization and expiration across the three physical configurations. Increasing the token lifetime consistently

improved token circulation by increasing token utilization and reducing token expiration. Across all three physical configurations, token utilization increased progressively from 65.8%-68.1% with a one-day lifetime to around 69.9%-71.9% after one week, further to 79.7%-82.4% after one month, and finally stabilized at 87.0%-92.0% when the lifetime reached three months. Meanwhile, the expiration rate decreased continuously from approximately 22.9%-26.1% to around 19.0%-22.3%, then to 9.0%-11.8%, and eventually approached zero after three months.

Beyond three months, both token utilization and expiration converged to stable values under all three physical configurations. As a result, the overall governance performance reached a plateau, with the overall score remaining essentially unchanged between three months and one year (0.661–0.662 for PV only, 0.672 for PV+BESS, and 0.671 for PV+BESS+BTM). This indicates diminishing operational benefits form further extending token validity under the current rule-based model. However, this should not be interpreted as a behaviorally optimal lifetime, since members do not adjust their redemption behavior according to the remaining validity period. In contrast, the fairness index remained nearly constant throughout the analysis (0.767 for PV only, 0.828 for PV+BESS, and 0.829 for PV+BESS+BTM), indicating that token lifetime primarily affects token circulation efficiency rather than the distribution of governance benefits among community members.

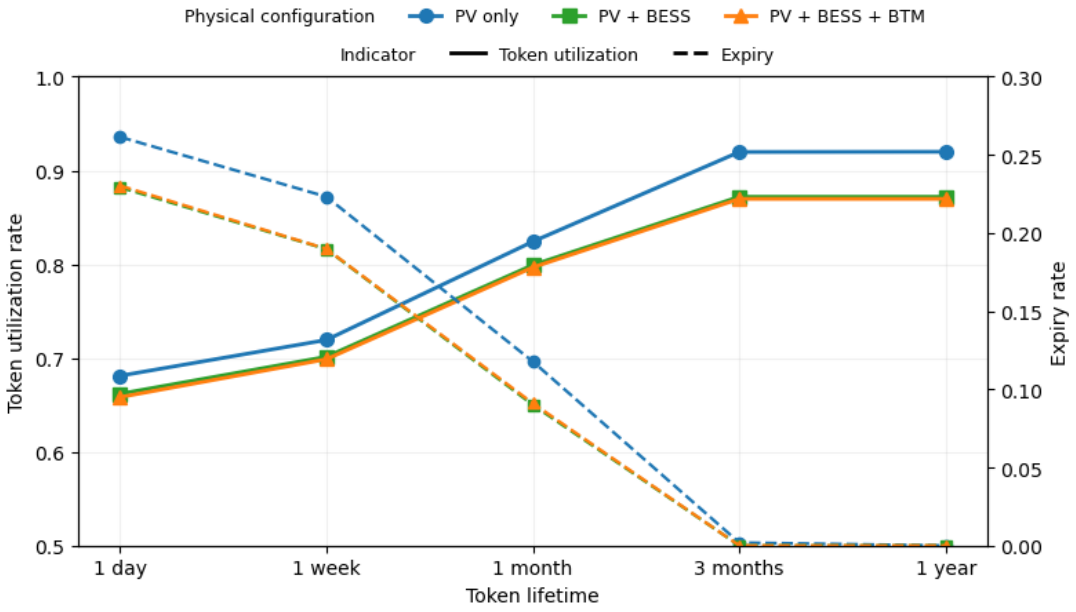


Figure 4.10: Token circulation under different token lifetimes

4.3.1.4 Demand Response Mode

Demand response (DR) incentives represent the primary expenditure channel of the community treasury in the proposed governance framework. Three governance

strategies were evaluated, namely no DR, peak reduction, and surplus absorption, to examine how different treasury-funded incentives influence governance performance and token circulation. For comparability, the same treasury allocation ratio was maintained across all DR modes. In the no-DR case, treasury tokens were still retained during the base allocation process, but no DR reward claims or payouts occurred, resulting in zero treasury utilization.

Figure 4.11 compares the overall governance performance under the three DR modes. Across all three physical configurations, the overall scores remained similar regardless of the selected DR mode. For the PV-only configuration, the overall score varied only between 0.618 and 0.620, while the corresponding ranges for the PV+BESS and PV+BESS+BTM configurations were 0.639-0.643 and 0.637-0.641, respectively. These results indicate that the introduction of treasury-funded demand response incentives does not substantially alter the integrated governance performance under the proposed evaluation framework.

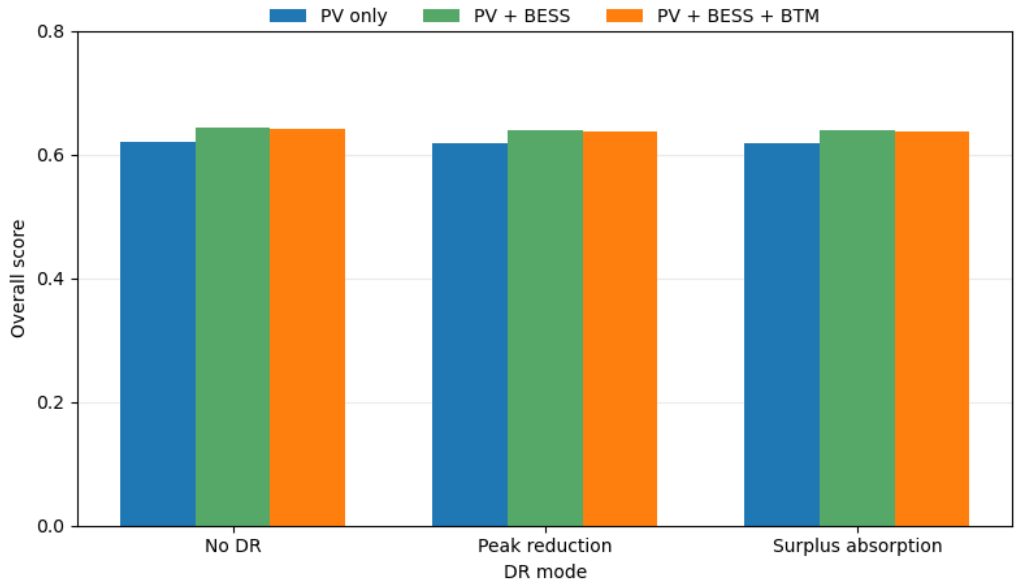


Figure 4.11: Effect of DR modes on overall governance performance

Figure 4.12 provides further insight into the underlying performance components. For clarity, the values shown represent the average across the three physical configurations. Compared with the no-DR case, both DR modes increased token activation by introducing treasury-funded demand-response incentives. Surplus absorption consistently achieved the highest activation score. For the PV-only configuration, activation score increased from 0.748 in the no-DR case to 0.794 under peak reduction and 0.840 under surplus absorption. Similar trends were observed under the PV+BESS and PV+BESS+BTM configurations. At the same time, treasury

utilization increased from zero in the no-DR case to approximately 0.443-0.717 under the two DR modes, confirming that treasury tokens were actively redistributed through demand-response incentives. However, these improvements were accompanied by a modest increase in token expiration and a gradual reduction in distributional fairness. Consequently, although treasury-funded demand-response incentives enhanced token circulation and treasury utilization, the associated trade-offs offset much of the activation gains in the overall governance evaluation.

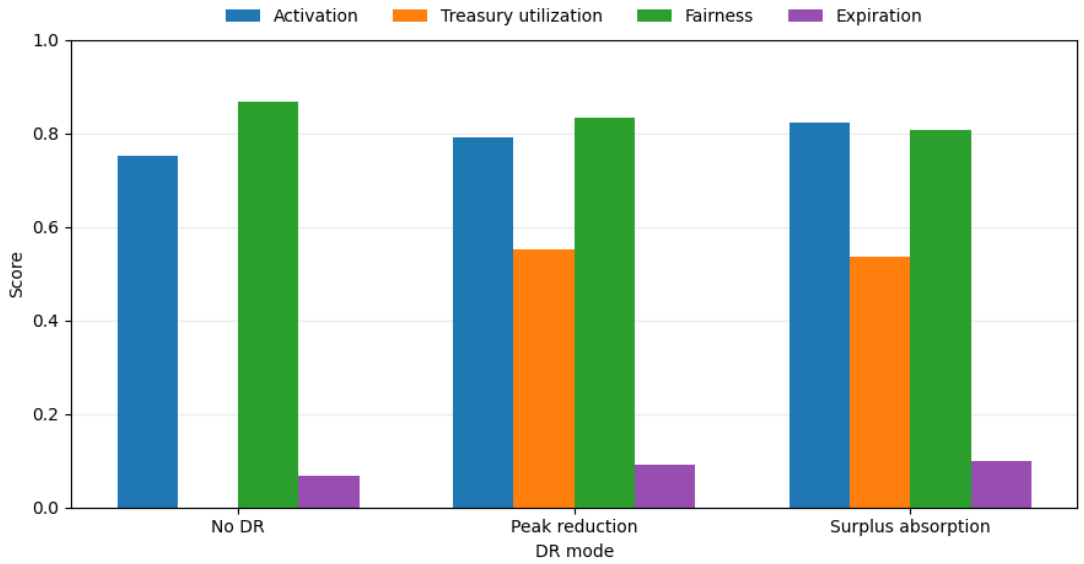


Figure 4.12: Performance components across DR modes

Figure 4.13 further visualized the activation-fairness trade-off across the three DR modes. The no-DR mode consistently occupies the high-fairness but low-activation region because the community treasury remains unused. In contrast, surplus absorption shifts the governance outcome toward higher activation at the expense of fairness, while peak reduction provides an intermediate balance between the two objectives. The consistent distribution of the three physical configurations indicates that the influence of DR mode is largely independent of the underlying energy system configuration. Under the prescribed DR-response assumptions used in the current model, these results suggest that the choice of DR mode mainly reflects different activation-fairness trade-offs rather than a universally superior governance strategy. In practice, behavioral responses to different DR incentives could alter the relative performance of the modes.

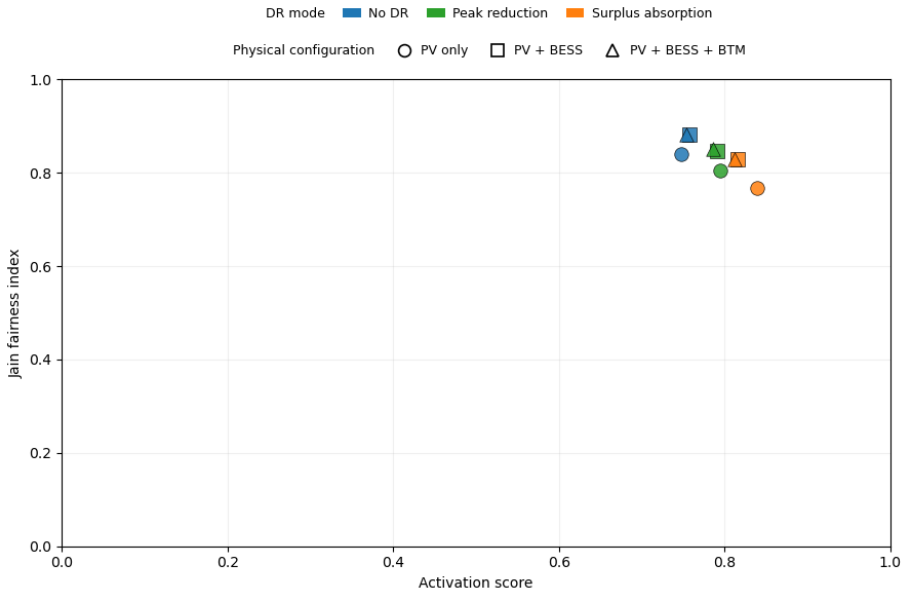


Figure 4.13: Activation-fairness trade-off across DR modes

4.3.1.5 Service spending ratio

The service spending ratio determines the proportion of each member's available token balance requested for community service redemption at each scheduled service spending event. Since member token balances accumulate over time while each token batch retains its own lifetime, increasing the service spending ratio increases the number of accumulated tokens redeemed through community services before expiration.

Figure 4.14 illustrates the influence of the service spending ratio on token utilization across the three physical configurations. Increasing the service spending ratio consistently improved token utilization while reducing token expiration. Under the PV-only configuration, the token utilization rate increased from 68.6% without service spending to 82.4% at a ratio of 0.10, and 91.8% at 0.20. The PV+BESS and PV+BESS+BTM configurations showed similar trends, where token utilization increased from approximately 66% to approximately 88% over the same range. Correspondingly, the expiration rate decreased from 25.6% to 2.4% for the PV-only configuration, from 22.2% to 1.3% for PV+BESS, and from 22.5% to 1.3% for PV+BESS+BTM. These results indicate that increasing the service spending ratio enables a larger proportion of accumulated member-held tokens to be redeemed through community services before reaching the end of their lifetime.

When the service spending ratio was increased beyond around 0.3, only marginal additional improvements were observed. Token utilization reached stable values, while the expiration rate approached zero under all three physical configurations. At the same time, the fairness index changed only slightly throughout the analysis, indicating

that the service spending ratio primarily influences token circulation efficiency rather than the distribution of governance benefits among community members. The highly consistent trends observed across the three physical configurations further suggest that the influence of service spending behavior is largely independent of the underlying physical energy configuration. Once the service spending ratio reaches approximately 0.30, most accumulated tokens that can realistically be redeemed before expiration have already been redeemed through the community service market, leaving little scope for further improvement.

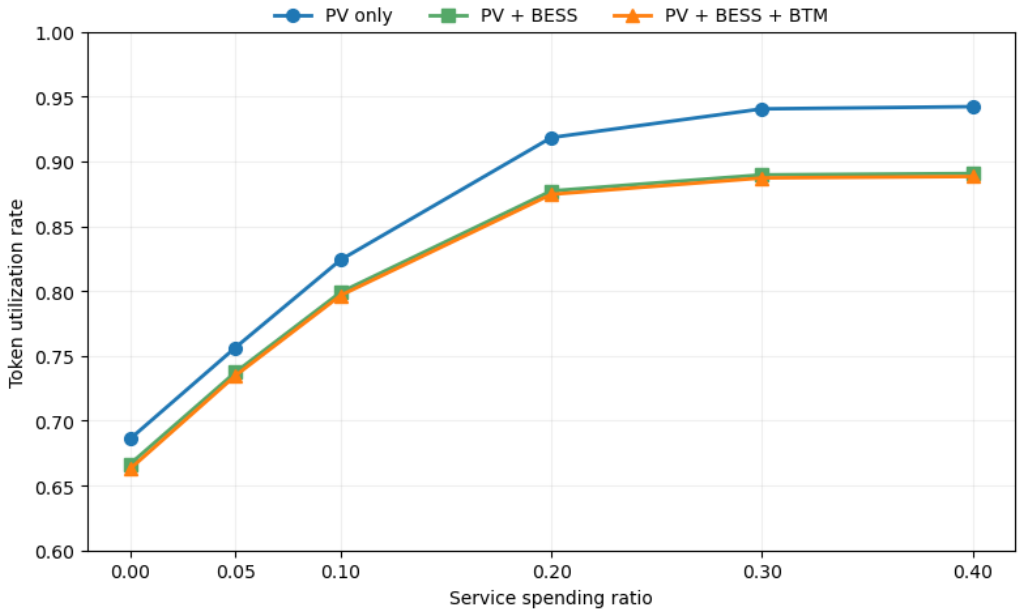


Figure 4.14: Token utilization under different service spending ratios

4.3.1.6 Service spending frequency

The service spending frequency determines how often members are able to redeem tokens for community services. Since token balances accumulate continuously while each token batch remains subject to a finite lifetime, more frequent service spending events provide additional opportunities for accumulated tokens to be redeemed before expiration.

Figure 4.15 presents the influence of service spending frequency on token utilization across the three physical configurations. Increasing the service spending frequency consistently improved token utilization while reducing token expiration. Under the PV-only configuration, the token utilization rate increased from 71.9% with monthly service spending to 82.4% with weekly spending, and further to 94.2% with daily spending. Hourly service spending provided only marginal improvements over the daily case. Similar trends were observed for the PV+BESS and PV+BESS+BTM configurations,

where token utilization increased from approximately 70% under monthly spending to around 89% under daily and hourly spending. Correspondingly, the expiration rate declined from around 19%-22% under monthly spending to around 9% under weekly spending and reached zero under both daily and hourly spending. These results indicate that increasing the service spending frequency enables accumulated member-held tokens to be redeemed more promptly before reaching the end of their lifetime.

Although hourly service spending achieved the highest overall governance performance, its advantage over daily spending was very small. The overall score increased by less than 0.01 across all three physical configurations, while both frequencies eliminated token expiration. These results suggest that increasing the service spending frequency beyond a daily schedule provides little additional governance benefit under the proposed governance framework.

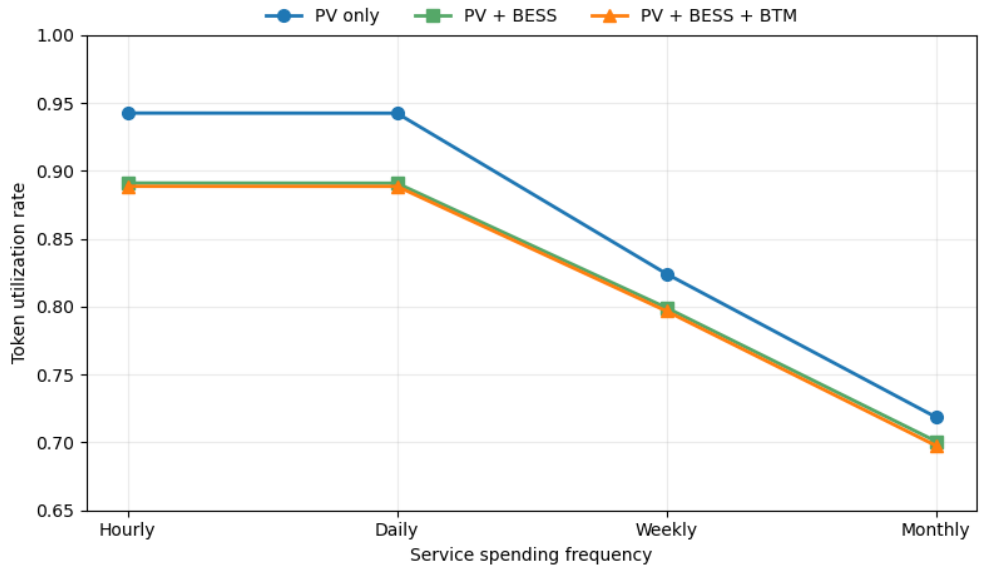


Figure 4.15: Token utilization under different service spending frequencies

4.3.2 Recommended Configuration

Following the parameter-specific design space exploration, an integrated governance assessment was conducted to identify the recommended governance configuration. Based on the findings of the parameter explorations presented in Section 4.3.1, the treasury allocation ratio, token lifetime, service spending ratio, and service spending frequency were refined to 10%, 90 days, 0.30, and daily, respectively, for the integrated assessment. The integrated candidate set therefore consisted of twelve governance designs formed by combining four allocation mechanisms with three DR modes under these refined parameter settings. Each candidate was subsequently evaluated under the three physical configurations, resulting in 36 integrated governance configurations.

Figure 4.16 presents the integrated governance design space in the activation-fairness plane. Each point represents one governance configuration generated by combining an allocation mechanism, a demand-response strategy, and a physical configuration under the refined governance parameter settings. The overall distribution remained broadly consistent across the three DR modes prior to the integrated ranking and final configuration selection.

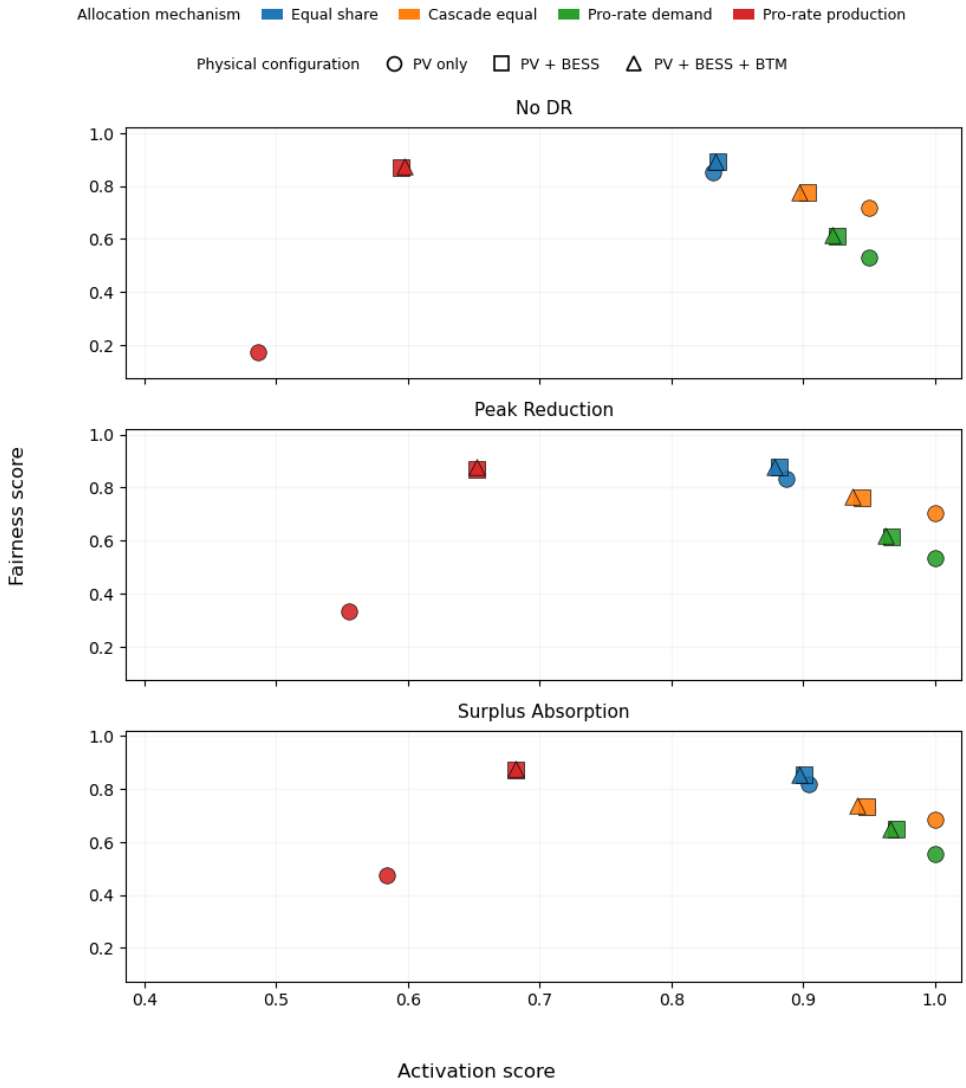


Figure 4.16: Integrated governance design space under different DR modes

Figure 4.17 ranks the ten highest-performing governance configurations according to the overall governance score. The highest-ranked configuration combined the PV+BESS physical configuration with equal share allocation and peak reduction demand response, achieving an overall governance score of 0.703. The remaining top-ranked configurations exhibited very similar performance, with overall scores differing by less than 0.002.

These results are consistent with the explorations presented in Section 4.3.1, where equal share and cascade equal share both demonstrated strong governance performance, while peak reduction and surplus absorption exhibited only marginal differences. The integrated assessment ultimately identified the combination of equal share and peak reduction as the highest-ranked governance configuration.

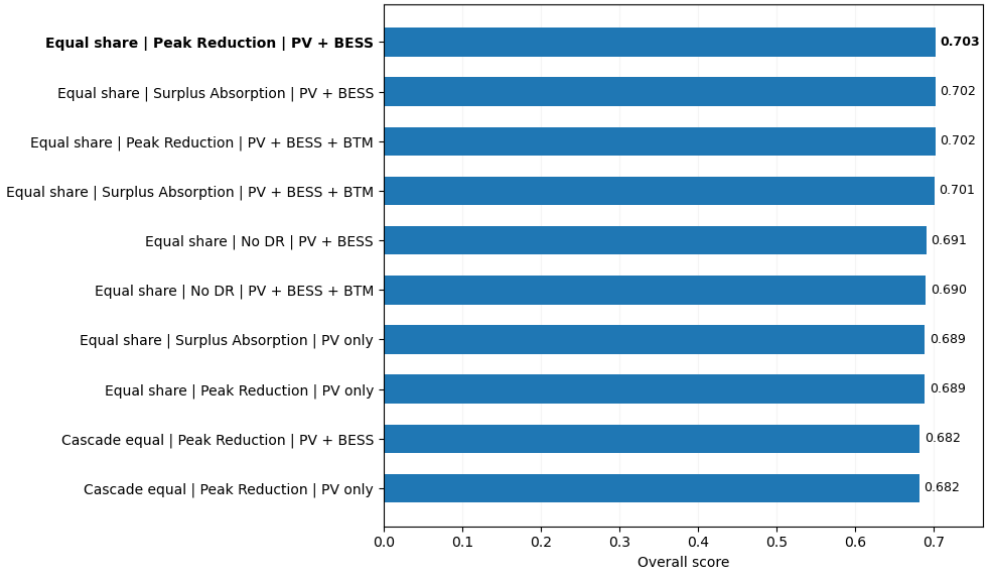


Figure 4.17: Top 10 integrated governance configurations

The recommended governance configuration is summarized in Table 4.2. This configuration was subsequently adopted for the governance-layer performance evaluation presented in the following section.

Table 4.2: Recommended governance configuration

Category	**Recommended**
Physical configuration	PV+BESS
Allocation mechanism	Equal share
Treasury allocation ratio	10%
Token lifetime	90 days
Demand response mode	Peak reduction
Service spending ratio	0.30
Service spending frequency	Daily

4.4 Governance Layer Performance

This section evaluates the performance of the selected governance design identified in Section 4.3. The analysis examines whether token mechanism was able to support effective circulation, broad participation, equitable value distribution, and sustained use of the community treasury. Performance is first assessed at the community level through aggregate governance indicators, annual token flows, monthly circulation dynamics, and token-enabled value creation. The analysis then proceeds to the member level to examine how token utilization and realized value were distributed across individual community participants.

4.4.1 Community-level Governance Performance

The overall governance performance of the selected token mechanism is summarized in Table 4.3. The integrated governance design achieved an overall governance score of 0.703, with an activation score of 0.881 and a fairness score of 0.875, indicating that the selected governance configuration maintained both high governance participation and a balanced distribution of governance benefits. The annual token utilization rate reached 98.3%, demonstrating that nearly all minted tokens were successfully redeemed through the consumption of energy access or community services. An energy access rate of 78.0% indicates that a substantial proportion of the eligible renewable surplus electricity was ultimately converted into token-enabled energy access. In addition, the treasury utilization rate reached 85.3%, suggesting that most treasury-held tokens were actively reinvested into community governance rather than remaining idle. No token expiration was observed during the simulation period, implying that the selected token lifetime provided sufficient circulation time for effective token utilization.

Table 4.3: Summary of performance indicators

Performance indicator	**Value**
Overall governance score	0.703
Activation score	0.881
Fairness score	0.875
Token utilization	98.3%
Energy access rate	78.0%
Treasury utilization	85.3%
Token expiration rate	0%

Figure 4.18 illustrates the annual circulation of tokens within the community. During the simulation year, 131.9k tokens were minted from eligible renewable electricity, of which 118.8k tokens were allocated directly to community members and 13.2k tokens were transferred to the community treasury. The treasury-held tokens were subsequently released through governance incentives, contributing 11.3k reward tokens to participating members, which then circulated alongside the tokens directly allocated to members. By the end of the year, only 326 tokens remained in member balances and 1.9k tokens remained in the treasury, consistent with the high overall token utilization reported in Table 4.3.

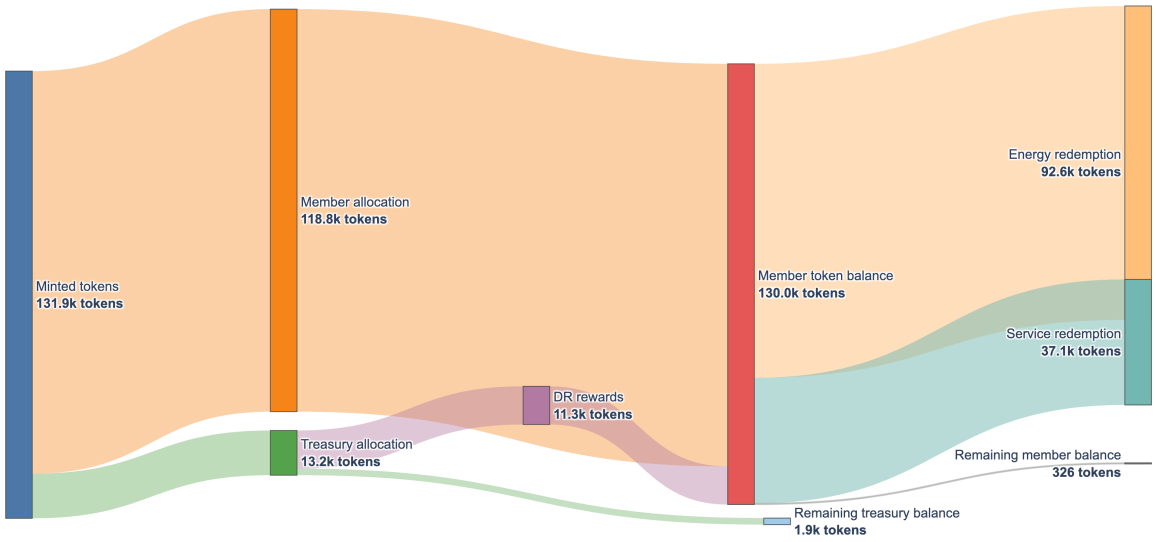


Figure 4.18: Annual community token flow

Figure 4.19 further presents the monthly governance dynamics throughout this simulation year, illustrating both the temporal pattern of token utilization and the corresponding evolution of member and treasury token balances. Energy redemption exhibited a pronounced seasonal pattern that broadly followed the monthly PV yield profile presented in Figure 4.1. Token utilization for energy increased rapidly from spring onwards as renewable electricity availability increased, reached its highest level in April, maintained a high level until September, and gradually declined towards winter. This behavior reflects the strong influence of seasonal renewable electricity availability on the energy-redemption pathway.

In contrast, service redemption exhibited a more gradual temporal pattern. Although service redemption remained relatively low during the first months of the year, it increased progressively from spring onwards and remained comparatively stable during the second half of the year. Unlike energy redemption, service redemption was determined by the configured service spending ratio applied to members' available

token balances. As token balances accumulated over time, service redemption increased accordingly and remained active beyond the seasonal peak in renewable electricity generation, contributing to sustained token circulation throughout the year.

The balance evolution further reflects these circulation dynamics. Member token balances remained low throughout the year, indicating that allocated tokens were generally redeemed shortly after being received rather than being accumulated over extended periods. Treasury balances increased gradually during the first three quarters as a portion of newly minted tokens was reserved for governance purposes, before declining towards the end of the year as treasury-held tokens were redistributed through demand response incentives. This seasonal pattern reflects the different dynamics of treasury inflows and outflows. Treasury inflows depend on token minting, which increases with renewable surplus generation, whereas treasury outflows are determined by DR reward payments. As PV surplus and token minting declined towards the end of the year while DR rewards continued to be paid, treasury outflows exceeded new inflows and the balance decreased. By the end of the simulation year, both member and treasury balances had decreased to relatively low levels, consistent with the high token and treasury utilization rates reported in Table 4.3.

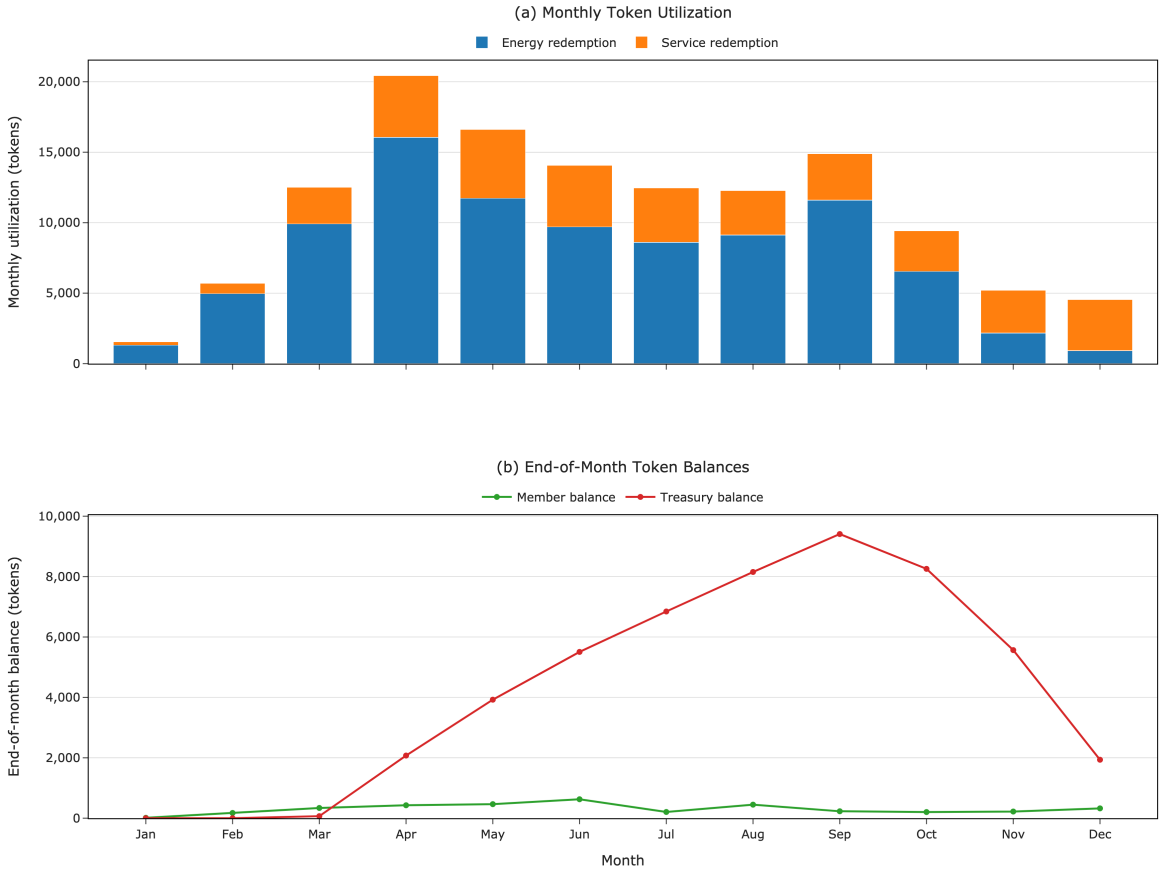


Figure 4.19: Monthly governance dynamics

Furthermore, Figure 4.20 presents the monthly community-level value realized through token utilization. Consistent with the token utilization dynamics, energy-related value exhibited a pronounced seasonal pattern, increasing during spring, remaining high throughout the summer months, and declining towards winter. In contrast, community service value was distributed more evenly over the year, reflecting the gradual increase in service redemption as token balances accumulated. As a result, community services accounted for a larger proportion of the total realized value during periods of lower renewable electricity generation. The proposed token system enabled locally generated renewable electricity to create both direct energy benefits and additional community service value, thereby extending the overall value generated within the energy community.

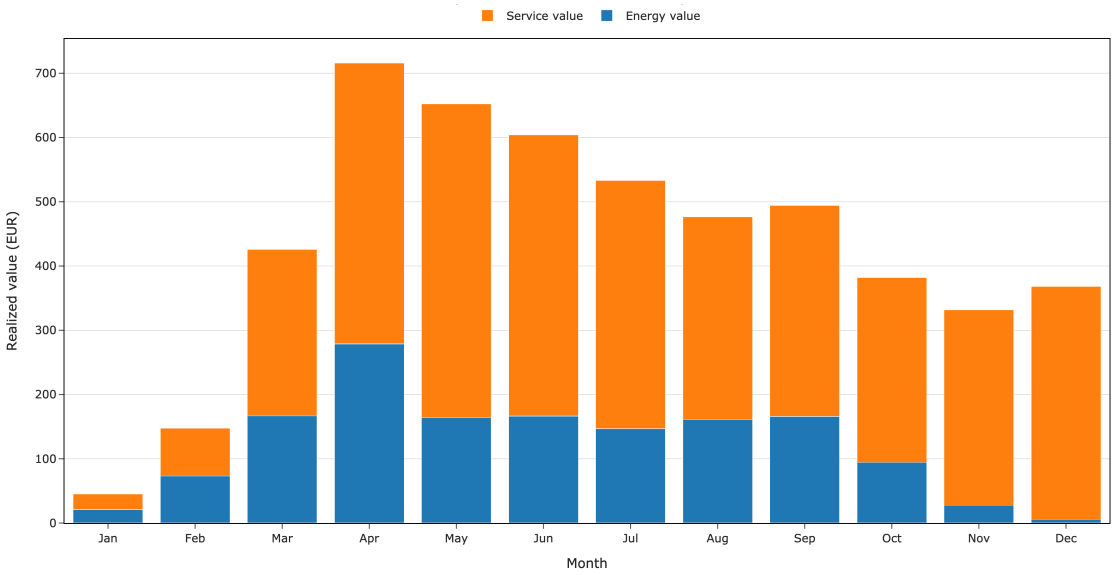


Figure 4.20: Monthly token-enabled community value

To examine the demand-side coordination generated by the token-enabled demand-response mechanism, the average community demand profiles before and after rule-based load shifting were compared, as shown in Figure 4.21. The peak reduction mechanism was activated during 876 hourly intervals in which the community grid import exceeded the 90th percentile of its annual distribution, corresponding to approximately 10% of the annual operating hours. As the proposed mechanism shifts rather than curtails flexible demand, the total annual community electricity demand remained unchanged while its temporal distribution was modified.

The proposed peak reduction mechanism consistently reduced community demand during the evening peak period, with the most noticeable reductions occurring between 17:00 and 20:00. The shifted demand was redistributed to adjacent non-peak hours,

resulting in a modest increase in demand immediately before and after the designated peak period while preserving the overall daily demand profile. These results show that, under the assumed flexibility and participation parameters, the proposed governance mechanism shifts demand away from identified peak periods without reducing total electricity consumption. The simulation illustrates how treasury-funded token incentives can support temporal demand coordination within the community.

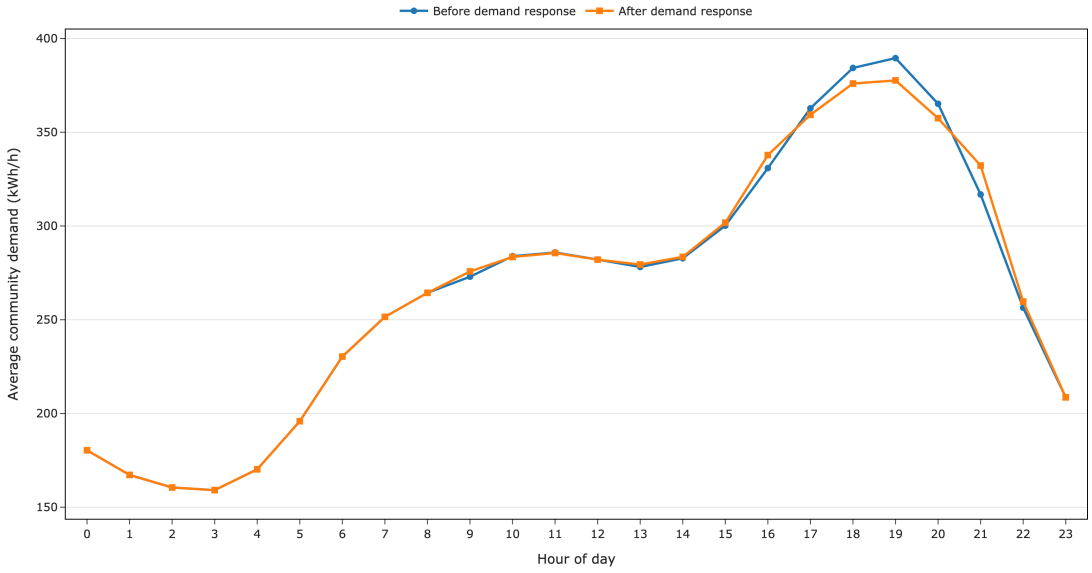


Figure 4.21: Average demand profile before and after demand response

Overall, the community-level results show that the proposed governance mechanism maintained effective token circulation, high utilization and sustained value creation throughout the simulation year. In addition to supporting token circulation and renewable energy value realization, the mechanism reshaped the simulated temporal demand profile while preserving overall electricity consumption. This demonstrates the potential for token-enabled governance to connect value distribution with demand-side coordination. However, these aggregate results do not reveal how the resulting benefits were distributed among individual community members. The following section therefore examines token utilization and realized value at the member level.

4.4.2 Member-level Governance Performance

The governance performance at the member level is presented by the annual composition of token utilization and the corresponding realized value for each community member, shown in Figure 4.22 and Figure 4.23. Although the overall

amount of redeemed tokens remained broadly comparable among members under the selected equal-share allocation mechanism, noticeable differences were observed in the relative contributions of energy redemption and service redemption.

As shown in Figure 4.22, members A, C, E, F, and I redeemed the majority of their tokens for energy access, whereas members D and J relied predominantly on community service redemption. Members G and H exhibited a more balanced utilization pattern between the two redemption pathways. These differences reflect the interaction between equal token allocation and heterogeneous member energy profiles. Members with larger residual demand and stronger temporal coincidence between their demand and available community renewable energy had more opportunities for energy redemption, whereas members with lower residual demand or fewer temporally matched redemption opportunities retained more tokens for subsequent service redemption.

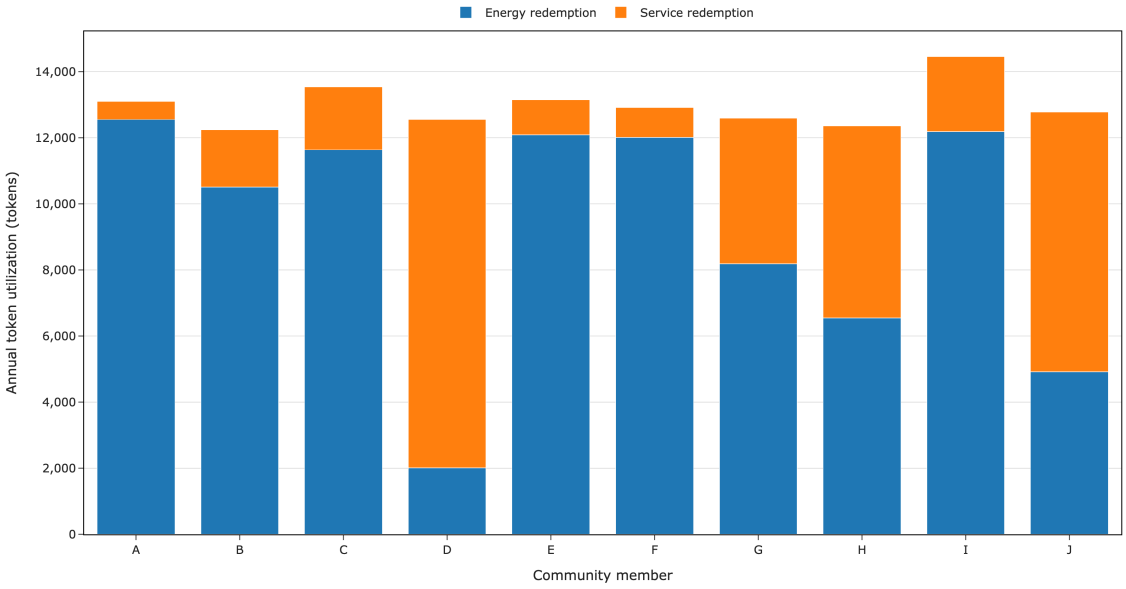


Figure 4.22: Member-level token utilization

These utilization patterns were directly reflected in the realized value distribution shown in Figure 4.23. Members whose token utilization was dominated by energy redemption obtained a larger proportion of their annual value through energy-related benefits, whereas members relying more heavily on service redemption derived a correspondingly greater share of their realized value from community services. Consequently, while the composition of realized value varied across members, the governance mechanism successfully enabled multiple pathways through which locally generated renewable electricity could be transformed into individual community benefits.

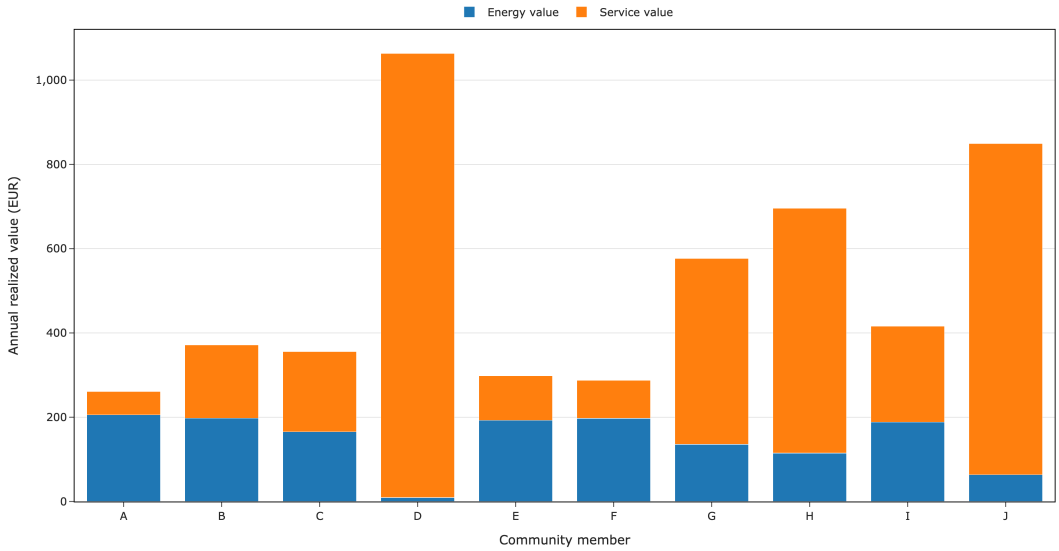


Figure 4.23: Member-level realized value

Overall, the member-level results demonstrate that the proposed governance mechanism not only maintained equitable opportunities for participation at the community level but also accommodated diverse utilization patterns among individual members, enabling renewable electricity to generate both energy-related and community-service value through a unified token framework.

5 Discussion

5.1 Principal Findings

The integrated evaluation provides several insights into the design of governance mechanisms for renewable energy communities. Rather than evaluating individual governance components in isolation, the proposed framework enables the interaction between physical energy operation, token design, and governance performance to be examined systematically. Three principal findings emerge from the results.

Finding 1. Eligible renewable energy serves as a meaningful value anchor.

The results indicate that eligible renewable energy, rather than total renewable electricity generation, provides a very meaningful value anchor for token issuance. In the proposed mechanism, only renewable electricity that has directly contributed to local community utilization, through immediate energy sharing or PV-origin battery discharge, is eligible for token issuance. This distinction separates renewable energy generation from renewable energy flows that contribute to local utilization, ensuring that token issuance reflects renewable energy consumed within the community rather than broader renewable electricity generation.

This value anchor has important implications for governance design. Because eligible renewable energy is determined by the underlying energy flows of the physical system, governance incentives become inherently coupled with physical system operation. Therefore, token issuance reflects not only the governance design itself but also the physical conditions that enable local renewable energy utilization. As a result, the token mechanism functions not merely as a reward for renewable generation, but as a governance instrument that links physically verified local energy contributions with incentives supporting collective community outcomes.

Finding 2. No single metric defines effective governance.

The results demonstrate that no single performance metric is sufficient to evaluate token-based governance. Across the evaluated configurations, improvements in one governance dimension were frequently accompanied by compromises in others. For example, allocation mechanisms that achieved higher activation generally produced lower distributional fairness, while demand-response incentives increased activation and treasury utilization but also increased token expiration and reduced fairness.

These trade-offs suggest that governance effectiveness should be understood as a multidimensional concept rather than being defined by a single objective. Therefore, governance evaluation requires the simultaneous consideration of multiple

performance dimensions that reflect different community priorities. This also explains why an integrated evaluation framework is necessary, as the preferred governance design ultimately depends on the objectives that a renewable energy community seeks to prioritize.

Finding 3. Effective governance requires coordinated token circulation.

According to the results, effective governance depends on the coordinated management of token circulation. Treasury allocation, token lifetime, service spending ratio, and service spending frequency collectively determine how long tokens remain available, how effectively they are utilized, and how community treasury resources are redistributed. Across the evaluated governance configurations, inappropriate parameter settings, such as short token lifetimes or excessive treasury allocation, resulted in lower token utilization or underutilized governance resources.

Within the investigated case study, the integrated evaluation identified the combination of PV+BESS, equal-share allocation, a 10% treasury allocation, a 90-day token lifetime, peak-reduction demand response, a service spending ratio of 0.30, and daily service spending as the highest-ranked configuration.

These findings suggest that effective token governance should not rely on any single circulation parameter independently. Instead, governance mechanisms should maintain sufficient token availability to support community coordination while ensuring that tokens continue to circulate and generate tangible value through renewable energy access and community services. Given the small performance differences among the highest-ranked alternatives, the selected configuration should be interpreted as a balanced case-specific design rather than a universally optimal solution. From this perspective, coordinated token circulation represents a fundamental design principle for sustaining effective governance performance in renewable energy communities.

5.2 Implications

The findings of this study provide several implications for the design of token-based governance mechanisms in renewable energy communities. Rather than seeking a universally optimal token mechanism, the proposed framework demonstrates a systematic approach for developing and comparing alternative governance designs under realistic community operating conditions. Collectively, these implications suggest that effective token design should be considered as a structured governance system in which multiple design components are coordinated to support community objectives.

Firstly, token mechanisms should be designed as coherent governance configurations rather than collections of independent design parameters. In this study, token issuance, allocation, treasury management, circulation, and demand response operate as interrelated components whose combined effects determine overall governance performance. Consequently, modifying individual parameters in isolation is unlikely to produce desirable governance outcomes. Instead, governance components should be configured collectively so that their interactions support the intended objectives of the community.

Secondly, the governance architecture of energy tokens is an important concern. The proposed two-pool structure separates newly issued tokens into an individual token pool for member-level energy access and a treasury token pool for community-level incentives. This separation enables individual participation and collective governance functions to coexist within the same token system while maintaining distinct purposes for each token pool. Such an architecture also provides flexibility for supporting community services and collective incentive programmes without fundamentally changing the underlying token issuance mechanism.

Thirdly, the study highlights the importance of systematic evaluation as part of the token design process. Because governance performance emerges from the interaction of multiple design components and multiple governance objectives, token mechanisms should be assessed using a multidimensional evaluation framework rather than individual performance indicators alone. Such an approach enables alternative governance configurations to be compared transparently and provides quantitative evidence to support token design decisions before practical implementation.

Finally, regulatory feasibility should be considered as an integral part of token design. The regulatory analysis indicates that energy-token mechanisms should remain closely connected to legally recognized energy-community activities and prioritize community participation, coordination, and service-oriented functions rather than direct payment, monetary settlement, or speculative exchange. In the Swedish context, where energy community activities continue to operate within the existing regulated electricity market structure, tokens are more appropriately positioned as a virtual coordination layer rather than as instruments representing the physical transfer or ownership of electricity. These considerations suggest that regulatory requirements can be incorporated as design boundaries from the outset, helping to define governance mechanisms that remain compatible with the broader institutional context of energy communities.

Beyond these governance and regulatory implications, the proposed framework also raises broader considerations regarding the environmental, social, and economic sustainability of renewable energy communities, which are discussed in the following section.

5.3 Sustainability Considerations

Sustainability represents an important broader consideration for the development of renewable energy communities. In the context of this study, sustainability is considered from environmental, social, and economic perspectives, while also recognizing the relevance of intra-generational and inter-generational equity in framing sustainability-related outcomes [59]. These perspectives are reflected in how the proposed physical-governance framework may support renewable energy utilization, equitable participation, and long-term economic performance. The purpose of this section is not to provide a comprehensive sustainability assessment, but to interpret the principal outcomes of the study from these complementary perspectives.

From an environmental perspective, the proposed framework supports more coordinated utilization of locally generated renewable electricity through energy sharing, battery storage, and demand-side coordination. The physical-layer results show that battery integration increases community self-consumption and self-sufficiency while reducing dependence on external grid exchange. At the governance layer, token issuance is specifically anchored to renewable electricity that contributes to community-level utilization through immediate energy sharing or PV-origin battery discharge. The demand-response mechanism further demonstrates the potential to shift electricity consumption away from identified peak periods without changing total annual demand. Together, these results show how the framework can support more effective local utilization and coordination of renewable energy resources. Environmental impacts such as lifecycle greenhouse-gas emissions and embodied impacts of PV and battery technologies were not explicitly quantified and therefore remain beyond the scope of the present assessment.

From a social perspective, the governance framework contributes to sustainability by explicitly addressing participation, distributional fairness, and access to shared community benefits. Alternative token allocation and circulation mechanisms were evaluated according to their influence on activation and fairness, while the selected governance configuration maintained relatively strong performance across both dimensions. The two-pool governance structure further allows individual token utilization and community-level treasury functions to coexist within the same mechanism. At the member level, participants were able to realize value through different combinations of renewable-energy access and community-service redemption, reflecting heterogeneous physical conditions and redemption opportunities. These characteristics are particularly relevant to intra-generational equity, since the framework explicitly considers how governance rules influence the distribution and realization of benefits among current community members. By making these distributional effects measurable, the framework provides a practical basis for designing more inclusive and transparent benefit-sharing arrangements within energy communities.

From an economic perspective, the physical-layer analysis demonstrates the importance of evaluating long-term system performance together with short-term operational benefits. Battery integration improved renewable-energy utilization and annual electricity-bill savings, but the additional capital investment and battery replacement costs substantially reduced lifecycle economic performance compared with the PV-only configuration under the investigated assumptions. The techno-economic layer therefore provides a practical basis for assessing whether improvements in renewable-energy utilization justify the additional investment associated with different physical configurations. At the governance layer, token circulation further creates complementary pathways for realizing community value through renewable-energy access and community services. Taken together, the framework supports lifecycle-informed decision-making by linking operational performance, investment requirements, and governance outcomes rather than evaluating these dimensions independently.

Overall, the proposed framework supports more sustainable energy-community design by bringing physical renewable-energy utilization, distributional governance, and lifecycle economic performance into a common analytical structure. The results demonstrate that these dimensions are closely interconnected: increasing renewable-energy utilization does not necessarily lead to the strongest lifecycle economic outcome, while governance strategies that improve token activation may introduce trade-offs in distributional fairness. By making these interactions explicit, the framework provides a structured basis for energy communities to identify configurations that better balance environmental, social, and economic objectives rather than optimizing any single dimension in isolation. Inter-generational considerations are partially represented through the 30-year lifecycle assessment, including battery degradation and replacement, while the governance analysis primarily addresses intra-generational distribution of benefits among current community members.

5.4 Limitations

Despite the contributions, several limitations should be acknowledged.

First, the proposed framework adopts a sequential integration between the physical and governance layers. The governance model is developed based on the operational outputs generated by the physical-layer simulation, while governance-induced behavioral changes are represented through rule-based modifications to energy consumption using predefined participation and flexibility assumptions rather than dynamically modelled member responses. However, these behavioral changes do not feed back into the physical dispatch optimization, meaning that battery operation is not re-optimized under updated demand conditions. Consequently, the current framework

represents an integrated design and evaluation approach rather than a fully closed-loop optimization framework. Accordingly, the demand-response analysis is interpreted as an assessment of governance-enabled load-shifting potential, while the resulting changes in battery dispatch and grid exchange remain a direction for further model integration.

Second, several governance functions are intentionally represented at an aggregated level. The service market is implemented as an aggregate service-credit channel through which members redeem a configurable proportion of their available tokens at predefined spending intervals. Token redemption is subject to an aggregate service-capacity ratio, and redeemed tokens are converted into service credits using a common conversion rate. However, the model does not distinguish between specific services or explicitly represent individual service providers, differentiated service prices, service-specific capacity constraints, or matching between service demand and supply. The euro-equivalent service value is included solely as a reference for evaluation and does not represent actual service pricing. Similarly, demand response events are identified using quantile-based thresholds, while flexible demand is represented as a configurable proportion of household electricity consumption together with predefined participation ratios. Consequently, individual flexible appliances and their operational characteristics, including appliance-specific power levels, flexibility windows, and operating constraints, are not explicitly represented within the current framework. The load-shifting procedure applies a standardized daily flexibility window and selects the nearest eligible non-event hour. Household-specific preferences, receiving-hour capacity, and potential rebound effects are not explicitly represented. Demand-response events are identified from the simulated annual energy-flow profiles and are therefore used for ex-post governance evaluation rather than operational event forecasting.

Third, the proposed token mechanism is developed using a rule-based governance design. Governance parameters, including token allocation, treasury allocation, token lifetime, and demand response incentives, are predefined and evaluated through scenario-based simulations rather than being determined through optimization. This modelling approach provides a transparent and interpretable basis for analyzing governance behavior but does not aim to identify a mathematically global optimum beyond the predefined governance design space. Furthermore, several governance rules involve discrete and non-continuous decision processes, which are not explicitly addressed within the present framework.

Fourth, the integrated governance score is based on a predefined set of indicators and weighting assumptions. Activation and fairness are assigned equal positive weights, while token expiration is treated as a penalty. These weights are normative design choices rather than empirically calibrated stakeholder preferences, and alternative weighting structures could change the ranking of governance configurations. In

addition, economic performance is not directly included in the governance score. Economic indicators are assessed separately at the physical-system level, while the current sequential framework does not quantify how different governance designs would feed back into physical operation and aggregate economic outcomes. A more comprehensive future assessment could therefore integrate governance, behavioral, and economic indicators within a common multi-criteria evaluation framework.

Finally, the proposed framework is demonstrated using a single residential renewable energy community under Swedish operating conditions. Although the modelling framework is designed to be transferable, its performance has not yet been evaluated across communities with different climatic conditions, regulatory environments, technology portfolios, or community structures. The conclusions of this study should therefore be interpreted within the context of the investigated case study.

5.5 Future Research

Targeting at the limitations of the present study, several promising directions can be identified for future research on token-based governance in renewable energy communities.

First, a fully integrated physical-governance optimization framework could be designed to provide a more comprehensive representation of the interactions between technical operation and community governance. In the current framework, governance-induced behavioral changes are not iteratively incorporated into the physical dispatch optimization. A bidirectional optimization framework could enable governance decisions and physical operation to influence one another dynamically, allowing energy system operation and governance mechanisms to be jointly optimized.

Second, future research could incorporate more detailed representations of community participation and local service ecosystems. Rather than modelling demand response and service markets at an aggregated level, future studies could explicitly represent appliance-level flexible loads, heterogeneous user preferences, and detailed community service markets involving multiple service providers, differentiated services, pricing mechanisms, and service capacities. These extensions would enable the framework to capture more realistic behavioral responses and community interactions while supporting the design of governance mechanisms that better reflect practical renewable energy communities.

Third, optimization-based and adaptive governance design could also be investigated. The rule-based governance mechanism developed in this study provides a transparent and interpretable foundation for analyzing governance behavior. However, governance design could be further formulated as an optimization problem in which governance parameters are automatically determined according to explicitly defined community

objectives. Because governance mechanisms often involve discrete decision rules, threshold-based policies, and multiple competing objectives, future research may require advanced optimization techniques or adaptive learning approaches capable of handling these characteristics. Such developments could enable governance mechanisms to evolve dynamically in response to changing operational conditions and community preferences.

Finally, broader validation across diverse renewable energy communities could further strengthen the applicability of the proposed framework. Further studies could evaluate the framework under different climatic conditions, regulatory environments, technology portfolios, and community scales, as well as investigate its implementation in pilot projects or living laboratories. Future sustainability analysis could also be extended by incorporating lifecycle environmental impacts, carbon-emission effects, and broader social indicators such as energy poverty, participation, and community acceptance. Such validation would provide valuable evidence regarding the transferability and practical deployment of token-based governance mechanisms across different renewable energy community contexts.

6 Conclusion

This thesis investigated how energy tokens can be designed to support governance within renewable energy communities through a physically grounded approach. To address this objective, an integrated framework was developed to link physical energy-system modelling with governance-layer token design. The comparative review of energy-community models, energy-token initiatives, and regulatory approaches provided the analytical basis for identifying the integration gaps addressed by the framework. A residential renewable energy community in Sweden was used as a case study to demonstrate the application of the proposed approach under the investigated operational and regulatory conditions.

A central methodological contribution of this thesis is the design of an integrated four-layer framework combining energy-system operation, tokenization, behavioral coordination, and token-value realization. The Energy System Layer models PV generation, battery storage, renewable-energy sharing, grid exchange, and lifecycle economic performance, thereby establishing the physical foundation for governance design. Based on the resulting energy flows, the Tokenization Layer identifies eligible renewable energy and connects it to token issuance and allocation. The Coordination Layer incorporates token circulation, treasury management, service utilization, and rule-based demand-response incentives, while the Value Realization Layer evaluates how redeemed tokens generate energy-related and community-service benefits. A multidimensional evaluation framework further enables alternative physical and governance configurations to be compared from technical, economic, participation, circulation, and distributional perspectives.

The quantitative results demonstrate the importance of considering physical, economic, and governance performance jointly. Compared with the PV-only configuration, the integration of community battery storage increased the self-consumption ratio from 59.8% to 64.9% and the self-sufficiency ratio from 25.7% to 27.9%, while reducing annual grid exports from 401.9 MWh to 347.0 MWh. The PV+BESS configuration achieved annual electricity-bill savings of EUR 122,126, compared with EUR 118,480 for PV only. However, the additional battery investment and replacement costs reduced the 30-year NPV from EUR 839,077 for PV only to EUR 215,889 for PV+BESS, demonstrating that improved annual operation does not necessarily result in stronger lifecycle economic performance.

At the governance level, the integrated evaluation identified a case-specific preferred configuration combining PV+BESS, equal-share allocation, a 10% treasury allocation, a 90-day token lifetime, peak-reduction demand response, a service spending ratio of 0.30, and daily service spending. This configuration achieved an overall governance score of 0.703, with an activation score of 0.881 and a fairness score of 0.875. The annual token utilization rate reached 98.3%, while the token-mediated energy access

rate and treasury utilization rate reached 78.0% and 85.3%, respectively. No token expiration occurred during the simulation year. These results indicate that high token circulation and substantial renewable-energy access can be achieved while maintaining a relatively balanced distribution of realized token-mediated energy access among community members.

The results further show that eligible renewable energy can serve as a meaningful value anchor for token issuance. By limiting issuance to immediate internal sharing and verified PV-origin BESS discharge, the framework links token creation to renewable energy that contributes to community utilization rather than to total renewable generation. Governance performance was nevertheless characterized by trade-offs rather than a single optimal criterion. Allocation mechanisms that increased activation did not always provide the strongest distributional equality, while treasury allocation, token lifetime, service utilization, and demand-response incentives interacted in determining circulation performance. Effective governance therefore depends on coordinated combinations of token-design parameters rather than on isolated mechanisms. The selected configuration should consequently be interpreted as a balanced solution for the investigated case, rather than as a universally optimal token design.

Overall, this thesis contributes a structured methodology for the systematic design and evaluation of energy-token mechanisms in renewable energy communities. The framework was designed specifically to connect physically verified renewable-energy flows with token issuance, allocation, circulation, demand-side coordination, and value realization, rather than applying tokens as an independent financial or settlement layer. From a broader sustainability perspective, it enables renewable-energy utilization, the distribution of community benefits, and long-term economic performance to be considered jointly within energy-community governance design. The study further demonstrates that the feasible design space is shaped by the EU and Swedish regulatory context. By positioning the proposed token as a physically grounded, utility-oriented, and community-internal coordination mechanism, the framework illustrates how regulatory requirements can be incorporated as design boundaries rather than treated solely as external compliance constraints. More broadly, integrating physical energy-system modelling with governance design provides a feasible pathway towards the systematic development of energy-token mechanisms for future renewable energy communities and other decentralized energy systems.

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Declaration on the Use of Artificial Intelligence

Generative AI tools were used during the preparation of this thesis to assist with language editing, programming support, and the generation of the thesis cover illustration.

In accordance with KTH's guidance on the responsible use of artificial intelligence, all research concepts, methodologies, data analysis, interpretation of results, and conclusions were independently developed, validated, and verified by the author. The author takes full responsibility for the accuracy, originality, and overall academic integrity of the work presented in this thesis.